R.K.D.F. UNIVERSITY, BHOPAL

B. Tech. (AGRICULTURAL ENGINEERING)

New Scheme Based On ICAR Flexible Curriculum

Semester – I

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Engineering Mathematics-I	BTA-5101	2-3 hr	3

Course Outcomes:

CO1	Know about matrices and their ranking
CO2	Understand the Eigen values, Eigen vectors and their properties
CO3	Distinguish between linear and orthogonal transformation
CO4	Develop knowledge on Cayley Hamilton theorem
CO5	Know about diagonalization of matrices
CO6	Understand the function of two or more independent variables
CO7	Know about Taylor's and Maclaurin's expansion
CO8	Distinguish between homogenous and composite function
CO9	Have a brief knowledge on Jacobian's error evaluation
CO10	Know about Stroke's divergence and Green's theorem

Course Contents:

UNIT–I

Matrices: Elementary transformations, Gauss-Jordon method to find inverse of a matrix, eigen values and Eigen vectors, Cayley-Hamilton theorem, linear transformation, diagonalization of matrices, Solution of linear equations, nature of rank, using Cayley-Hamilton theorem to find inverse of A.

UNIT–II

Differential calculus: Taylor's and Maclaurin's expansions; function of two or more independent variables, partial differentiation, homogeneous functions and Euler's theorem, composite functions, total derivatives, maxima and minima.

UNIT–III

Integral calculus: volumes and surfaces of revolution of curves; double and triple integrals, change of order of integration, application of double and triple integrals to find area and volume.

UNIT–IV

Vector calculus: Differentiation of vectors, scalar and vector point functions, vector differential operator Del, Gradient of a scalar point function, Divergence and Curl of a vector point function and their physical interpretations, identities involving Del, second order differential operator; line, surface and volume integrals, Stoke's, divergence and Green's theorems (without proofs).

References

1. Narayan Shanti. 2004. Differential Calculus. S. Chand and Co. Ltd. New Delhi.
2. Narayan Shanti. 2004. Integral Calculus. S. Chand and Co. Ltd. New Delhi.
3. Grewal B S. 2004. Higher Engineering Mathematics. Khanna Publishers Delhi.
4. Narayan Shanti. 2004. A Text Book of Vector. S. Chand and Co. Ltd. New Delhi.

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Semester – I

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Engineering Physics	BTA-5102	2-3 hr	3

Course Outcomes:

CO 1	Distinguish among conductor, insulator and semiconductors in terms of band gap
CO 2	Have a brief knowledge on semiconductors
CO 3	To know the quantum nature of electrons and effective mass of electrons and holes
CO 5	Know about different types of magnetic materials and their properties
CO 6	Understand the dual nature of particles
CO 7	Understand the effect of electric and magnetic field on spectra of atoms and molecules
CO 8	Know about super conducting materials and their properties
CO 9	Know about different types of LASER and MASERS
CO 10	Understand holography
CO 11	Know about optical fibres and its applications
CO 12	Know about illumination and its laws.

Course Contents:

UNIT–I

Dia, para and ferromagnetism-classification. Langevin theory of dia and paramagnetism. Adiabatic demagnetization. Weiss molecular field theory and ferromagnetism. Curie-Weiss law. Wave particle quality, de-Broglie concept, uncertainty principle.

UNIT–II

Wave function. Time dependent and time independent Schrodinger wave equation, Qualitative explanation of Zeeman effect, Stark effect and Paschan Back effect, Raman spectroscopy.

UNIT–III

Statement of Bloch's function. Bands in solids, velocity of Bloch's electron and effective mass. Distinction between metals. insulators and semiconductors. Intrinsic and extrinsic semiconductors, law of mass action. Determination of energy gap in semiconductors. Donors and acceptor levels.

UNIT–IV

Superconductivity, critical magnetic field. Meissner effect. Isotope effect. Type-I and II superconductors, Josephson's effect DC and AC, Squids. Introduction to high Tc superconductors.

Spontaneous and stimulated emission, Einstein A and B coefficients. Population inversion, He-Ne and Ruby lasers. Ammonia and Ruby masers, Holography-Note.

UNIT–V

Optical fiber. Physical structure. basic theory. Mode type, input output characteristics of optical fiber and applications. Illumination: laws of illumination, luminous flux, luminous intensity, candle power, brightness.

References

1. Brijlal and Subrahmanyam.Text Book of optics. S. Chand and Co., New Delhi.
2. Sarkar Subir Kumar. Optical State Physics and Fiber Optics. S. Chand and Co., New Delhi.
3. Gupta S L, Kumar V Sharma R C. Elements of Spectroscopy. Pragati Prakasam, Meeruth.
4. Saxena B S and Gupta R C. Solid State Physics. Pragati Prakasam, Meeruth.
5. Srivastava B N. Essentials of Quantum Mechanics. Pragati Prakasam, Meeruth.

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Semester – I

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Engineering Chemistry	BTA-5103	2-3 hr	3

Course Outcomes:

CO1 Know different type of fuel and their quality through various analytical methods
CO2 Understand the cause, effect and prevention of corrosion
CO3 Learn about the type of water and its effect on industries
CO4 Acquire knowledge about the modern instruments for analysis of samples
CO5 Know the nutrition value of food, types of food preservatives, colouring reagents and flavouring reagents
CO6 Acquire basic idea about polymerization and its properties
CO7 Determination of molecular mass through various methods

Course Contents:

UNIT–I Phase rule and its application to one and two component systems. **Fuels:** classification. Calorific value. **Colloids:** classification. Properties.**Corrosion:** causes, types and method of prevention.

UNIT–II Water: temporary and permanent hardness, Disadvantages of hard water, scale and sludge formation in boilers, boiler corrosion. Analytical methods like thermo-gravimetric. polarographic analysis. nuclear radiation.

UNIT–III Detectors and analytical applications of radioactive materials. Enzymes and their use in the manufacturing of ethanol and acetic acid by fermentation methods.

UNIT–IV Principles of food chemistry. Introduction to lipids, proteins, carbohydrates, vitamins, food preseltators, colouring and flavouring reagents of food. Lubricants: properties. Mechanism. Classification and tests.

UNIT–V Polymers. Types of polymerization. Properties. Uses and methods for the determination of molecular weight of polymers. Introduction to IR spectroscopy.

References

1. Organic Chemistry: Robert. T. Morrison and Robert. N .Boyd, Pearson publication
2. Organic Chemistry: I. L. Finar, Pearson publication
3. Spectroscopy: By Y.R.Sharam, Kalyani Publisher
4. Engg. Chemistry by Jain and Jain, Dhanpat Rai Publication.
5. Physical Chemistry by Samuel Glasstone by Mc. Graw Hill.

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Semester – I

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Principles of Soil Science	BTA-5104	2-3 hr	3

Course Outcomes:

CO1 Understand the origin of soil, soil forming rocks minerals, their classification and composition.

CO2 Know about the physical and chemical properties of soil.

CO3 Acquaint with the important of clay minerals & their role in nutrient transformation in soil,

CO4 Gain knowledge regarding the role of plant nutrients, deficiency symptoms, important fertilizer and their reaction in soil.

CO5 Know about the problematic soil, their characteristics and management.

Course Contents:

UNIT -1 Nature and origin of soil; soil forming rocks and minerals, their classification and composition, soil forming processes,

UNIT -II classification of soils – soil taxonomy orders. Important soil physical properties; and their importance; soil particle distribution; soil inorganic colloids – their composition,

UNIT -III properties and origin of charge; ion exchange in soil and nutrient availability; soil organic matter – its composition and decomposition, effect on soil fertility; soil reaction – acidic, saline and sodic soils; quality or irrigation water.

UNIT -IV Essential plants nutrients – their functions and deficiency symptoms in plants; important inorganic fertilizers and their reactions in soils.

UNIT -V Use of saline and sodic water for crop production, Gypsum requirement for reclamation of sodic soils and neutralizing RSC; Liquid fertilizers and their solubility and compatibility.

References

1. The nature and properties of soil (2002) - N.C. Brady and Ray, R. Weill; Pearson Education Inc. New Delhi.
2. Fundamentals of soil science (2002) - Indian Society of soil science, IARI, New Delhi.
3. Soil pedology (1996) – J. L. Sehgal, Kalyani publication, Ludhiana.
4. Introduction to soil physics-Daniel Hillel, Academic Press, New York

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Semester – I

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Surveying and Leveling	BTA-5105	2-3 hr	3

Course Outcomes:

CO1	Understand the importance of surveying, how to prepare maps, plans and the importance of different scales
CO2	Know about taking measurements linear, angular and altitude etc.
CO3	Know about the different land features, specifications of work and to give layout of Engineering structures
CO4	Computation of area of fields and volume of earthwork, items of works in construction

Course Contents:

UNIT – I

Surveying: introduction, classification and basic principles, linear measurements. Chain surveying. Cross staff survey, compass survey. Planimeter, errors in measurements, their elimination and correction. Plane table surveying.

UNIT – II

Levelling: leveling difficulties and error in leveling, contouring, computation of area and volume. Theodolite traversing.

UNIT – III

Introduction to setting of curves. Total station, electronic theodolite. Introduction to GPS survey.

References

1. Punmia, B.C. Surveying, Vol.I.Laxmi Publications, New Delhi
2. Agor, R. A Text Book of Surveying & Levelling, Khanna Publishers, New Delhi
3. Kanetkar, T.P.1993. Surveying and Levelling, Pune Vidyarthi Grihaprakashan, Pune

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Semester – I

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Engineering Mechanics	BTA-5106	2-3 hr	3

Course Outcomes:

- **CO1** Understand the importance of different force system, composition, resolution and analysis of force systems
- **CO2** Know about moments, couples, C.G., moment of inertia, FBD and frictional forces
- **CO3** Know about the different simple framed structures, shear force, stresses and torsion in the beams

Course Contents:

UNIT – I

Basic concepts of Engineering Mechanics. Force systems, Centroid, Moment of inertia, Free body diagram and equilibrium of forces. Frictional forces

UNIT – II

Analysis of simple framed structures using methods of joints, methods of sections and graphical method.

UNIT – III

Simple stresses. Shear force and bending moment diagrams. Stresses in beams, Torsion. Analysis of plane and complex stresses.

References

1. Sundarajan V 2002.Engineering Mechanics and Dynamics. Tata McGraw Hill Publishing Co. Ltd., New Delhi.
2. Timoshenko S and Young D H 2003.Engineering Mechanics. McGraw Hill Book Co., New Delhi.
3. Prasad I B 2004.Applied Mechanics.Khanna Publishers, New Delhi.
4. Prasad I B 2004.Applied Mechanics and Strength of Materials.Khanna Publishers, New Delhi.
5. Bansal R K 2005.A Text Book of Engineering Mechanics.Laxmi Publishers, New Delhi.

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Semester – I

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Engineering Drawing	BTA-5107	2-3 hr	2

Course Outcomes:

CO 1. Know the principle & construction of different types of scales, geometrical figures etc
CO 2. Understand the principle of isometric projection and able to draw the same
CO 3. Able to draw 1st& 3rd angle projection of points , lines and solids
CO 4. Understand the principle of development of surfaces, section of solids and able to draw the same
CO 5. Understand the principle of conversion of pictorial view to orthographic view and able to draw the same
CO 6. Drawing of screw thread, nuts and bolts, welding symbol, rivet joint etc

Course Contents:

UNIT – I

Introduction of drawing scales; First and third angle methods of projection. Principles of orthographic projections;

UNIT – II

References planes; Points and lines in space and traces of lines and planes; Auxiliary planes and true shapes of oblique plain surface; True length and inclination of lines;

UNIT – III

Projections of solids (Change of position method, alteration of ground lines); Section of solids and Interpenetration of solid surfaces; Development of surfaces of geometrical solids;

UNIT – IV

Isometric projection of geometrical solids. Preparation of working drawing from models and isometric views. Drawing of missing views. Different methods of dimensioning. Concept of sectioning. Revolved and oblique sections. Sectional drawing of simple machine parts. Types of rivet heads and riveted joints. Processes for producing leak proof joints. Symbols for different types of welded joints. Nomenclature, thread profiles, multi start threads, left and right hand threads. Square headed and hexagonal nuts and bolts. Conventional representation of threads.

UNIT – V

Different types of lock nuts, studs, machine screws, cap screws and wood screws. Foundation bolts. Forms of screw threads like metric thread, whit worth thread, square thread, acme thread, knuckle thread, buttress thread etc., Bolts- headed centre, stud screws, set screws, butt, hexagonal and square; keys-types, taper, rank taper, hollow saddle etc.

References

1. Bhat N D. 2010. Elementary Engineering Drawing. Charotar Publishing House Pvt. Ltd., Anand.
2. Bhatt N D and Panchal V M. 2013. Machine Drawing. Charotar Publishing House Pvt. Ltd., Anand.
3. Narayana K L and Kannaiah P. 2010. Machine Drawing. Scitech Publications (India) Pvt. Ltd., Chennai.

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Semester – I

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Heat and Mass Transfer	BTA-5108	2-3 hr	2

Course Outcomes:

CO 1 Understand the principles of conduction, convection, radiation
CO 2 Analyze the mathematical and practical aspect of conduction, convection and radiation
CO 3 Analyze the mathematical and practical aspect of heat exchangers
CO 4 Understand the principles of mass transfer and related laws.

Course Contents:

UNIT – I

Concept, modes of heat transfer, thermal conductivity of materials, measurement. General differential equation of conduction. One dimensional steady state conduction through plane and composite walls, tubes and spheres with and without heat generation. Electrical analogy.

UNIT – II

Insulation materials. Fins, Free and forced convection. Newton's law of cooling, heat transfer coefficient in convection.

UNIT – III

Dimensional analysis of free and forced convection. Useful non dimensional numbers. Equation of laminar boundary layer on flat plate and in a tube. Laminar forced convection on a flat plate and in a tube. Combined free and forced convection.

UNIT – IV

Introduction. Absorptivity, reflectivity and transmissivity of radiation. Black body and monochromatic radiation, Planck's law, Stefan-Boltzman law, Kirchoff's law, grey bodies and emissive power, solid angle, intensity of radiation. Radiation exchange between black surfaces, geometric configuration factor. Heat transfer analysis involving conduction, convection and radiation by networks.

UNIT – V

Types of heat exchangers, fouling factor, log mean temperature difference, heat exchanger performance, transfer units. Heat exchanger analysis restricted to parallel and counter flow heat exchangers. Steady state molecular diffusion in fluids at rest and in laminar flow, Flick's law, mass transfer coefficients. Reynold's analogy.

References

1. Geankoplis C.J. 1978. Transport Port Processes and Unit Operations.Allyn and Bacon Inc., Newton, Massachusetts.
2. Holman J P. 1989. Heat Transfer. McGraw Hill Book Co., New Delhi.
3. Incropera F P and De Witt D P. 1980. Fundamentals of Heat and Mass Transfer. John Wiley and Sons, New York.
4. Gupta C P and Prakash R. 1994.Engineering Heat Transfer. Nem Chand and Bros., Roorkee.

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Semester – II

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Engineering Mathematics–II	BTA-5201	2-3 hr	3

Course Outcomes:

CO 1	Understand Cauchy's and Legendre's linear equations
CO 2	Know about series solution techniques
CO 3	Distinguish between the exact and Bernoulli's differential equations
CO 4	Apply one dimensional heat and wave equation
CO 5	Distinguish between various functions of complex variables
CO 6	Understand Cauchy Reimann equation
CO 7	Know about ratio and root test
CO 8	Know about formation of partial differential equation
CO 9	Know about Lagrange's linear equation and Charpit's method
CO 10	Know about even and odd functions and their fourier series

Course Contents:

UNIT – I

Ordinary differential equations: Exact and Bernoulli's differential equations, equations reducible to exact form by integrating factors, equations of first order and higher degree, Clairaut's equation,

UNIT – II

Differential equations of higher orders, methods of finding complementary functions and particular integrals, method of variation of parameters, Cauchy's and Legendre's linear equations, simultaneous linear differential equations with constant coefficients, series solution techniques, Bessel's and Legendre's differential equations.

UNIT – III

Functions of a Complex variable: Limit, continuity and analytic function, Cauchy-Riemann equations, Harmonic functions. Infinite series and its convergence, periodic functions, Fourier series, Euler's formulae, Dirichlet's conditions, functions having arbitrary period, even and odd functions, half range series, Harmonic analysis.

UNIT – IV

Fourier Sine and Cosine Series, Fourier series for function having period 2L, Elimination of one and two arbitrary function. Partial differential equations: Formation of partial differential equations

UNIT – V

Higher order linear partial differential equations with constant coefficients, solution of non-linear partial differential equations, Charpit's method, application of partial differential equations (one dimensional wave and heat flow equations, Laplace Equation.

References:

1. Narayan Shanti. 2004. A Text Book of Matrices. S. Chand and Co. Ltd. New Delhi.
2. Grewal B S. 2004. Higher Engineering Mathematics. Khanna Publishers Delhi.
3. Ramana B V. 2008. Engineering Mathematics. Tata McGraw-Hill. New Delhi.

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Semester – II

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Environmental Science and Disaster Management	BTA-5202	2-3 hr	3

Course Outcomes:

CO 1 Understand different resources such as mineral resources, food resources, water resource, energy resources, natural resources and land resources.

CO 2 Gather knowledge about environmental pollution, Soil pollution, Air pollution and thermal pollution

CO 3 Acquaint themselves with waste land reclamation, Ecosystems and their management

CO 4 Learn regarding Biodiversity and its conservation, natural disaster and manmade disaster along with their management

Course Contents:

UNIT – I

Environmental Studies: Scope and importance. Natural Resources: Renewable and non-renewable resources Natural resources and associated problems. a) Forest resources: Use and over-exploitation, deforestation, case studies. Timber extraction, mining, dams and their effects on forest and tribal people. b) Water resources: Use and over-utilization of surface and ground water, floods, drought, conflicts over water, dams-benefits and problems. c) Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies. d) Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies. e) Energy resources: Growing energy needs, renewable and non-renewable energy sources, use of alternate energy sources. Case studies. f) Land resources: Land as a resource, land degradation, man induced landslides, soil erosion and desertification. Role of an individual in conservation of natural resources. Equitable use of resources for sustainable lifestyles.

UNIT – II

Ecosystems: Concept, Structure, function, Producers, consumers, decomposers, Energy flow, ecological succession, food chains, food webs, ecological pyramids. Introduction, types, characteristic features, structure and function of the forest, grassland, desert and aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries).

UNIT – III Biodiversity and its conservation:- Introduction, definition, genetic, species & ecosystem diversity and bio-geographical classification of India. Value of biodiversity: consumptive use, productive use, social, ethical, aesthetic and option values. Biodiversity at global, National and local levels, India as a mega-diversity nation. Hot-sports of biodiversity. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts. Endangered and endemic species of India. Conservation of **biodiversity**: In-situ and Ex-situ conservation of biodiversity.

UNIT – IV

Environmental Pollution: definition, cause, effects and control measures of a.) Air pollution b.) Water pollution c.) Soil pollution d.) Marine pollution e.) Noise pollution f.) Thermal pollution g.) Nuclear hazards. Solid Waste Management: causes, effects and control measures of urban and industrial wastes. Role of an individual in prevention of pollution. Pollution case studies. Social Issues and the Environment from Unsustainable to Sustainable development, Urban problems related to energy. Water conservation, rain water harvesting, watershed management. Environmental ethics: Issues and possible solutions, climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. dies. Wasteland reclamation. Consumerism and waste products. Environment Protection Act. Air (Prevention and Control of Pollution) Act. Water (Prevention and control of Pollution) Act. Wildlife Protection Act. Forest Conservation Act. Issues involved in enforcement of environmental legislation. Public awareness. Human Population and the Environment: population growth, variation among nations, population explosion, Family Welfare Programme. Environment and human health: Human Rights, Value Education, HIV/AIDS. Women and Child Welfare. Role of Information Technology in Environment and human health.

UNIT – V

Disaster Management:

Natural Disasters and nature of natural disasters, their types and effects. Floods, drought, cyclone, earthquakes, landslides, avalanches, volcanic eruptions, Heat and cold waves, Climatic change: global warming, Sea level rise, ozone depletion. Man Made Disasters-Nuclear disasters, chemical disasters, biological disasters, building fire, coal fire, forest fire, oil fire, air pollution, water pollution, deforestation, industrial waste water pollution, road accidents, rail accidents, air accidents, sea accidents. Disaster Management- Effect to migrate natural disaster at national and global levels. International strategy for disaster reduction. Concept of disaster management, national disaster management framework; financial arrangements; role of NGOs, community-based organizations and media. Central, state, district and local administration; Armed forces in disaster response; Disaster response; Police and other organizations.

References:

1. Bharucha Erach. 2005. Text Book of Environmental Studies for Undergraduate Courses. University Grants Commission, University Press, Hyderabad.
2. Sharma J P. 2003. Introduction to Environment Science. Lakshmi Publications.
3. Chary Manohar and Jaya Ram Reddy. 2004. Principles of Environmental Studies. B.S Publishers, Hyderabad.

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Semester – II

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Entrepreneurship Development & Business Management	BTA-5203	2-3 hr	3

Course Outcomes:

CO 1	Basic concept of entrepreneur
CO 2	Entrepreneurship development process
CO 3	Skill required becoming an entrepreneur
CO 4	Analysis of internal and external environment for business opportunity
CO 5	Government policy available for entrepreneurs
CO 6	Business communication skill
CO 7	Managerial skill required to become an entrepreneur
CO 8	Problem solving skill
CO 9	Quality management
CO 10	Project preparation techniques
CO 11	Process to set up new enterprises

Course contents:

Unit-I

Entrepreneurship , management – Management functions – planning- Organizing -Directing – motivation – ordering – leading – supervision-Communication and control – Capital – Financial management – importance of financial statements – balance sheet – profit and loss statement, Analysis of financial statements – liquidity ratios – leverage ratios, Coverage ratios– turnover ratios – profitability ratios, Agro-based industries –

Unit-II

Project – project cycle – Project appraisal and evaluation techniques – undiscounted measures – payback period – proceeds per rupee of outlay, Discounted measures – Net Present Value (NPV) – Benefit-Cost Ratio (BCR) – Internal Rate of Return (IRR) – Net benefit investment ratio (N / K ratio) – sensitivity analysis-Importance of agribusiness in Indian economy International trade-WTO agreements – Provisions related to agreements in agricultural and food commodities.

Unit-III

Agreements on agriculture (AOA) – Domestic supply, market access, export subsidies agreements on sanitary and phyto-sanitary (SPS) measures, Trade related intellectual property rights (TRIPS). Development (ED): Concept of entrepreneur and entrepreneurship Assessing overall business environment in Indian economy– Entrepreneurial and managerial characteristics- Entrepreneurship development Programmes (EDP)- Generation incubation and commercialization of ideas and innovations- Motivation and entrepreneurship development- Globalization and the emerging business entrepreneurial environment-Managing an enterprise:

Unit-IV

Importance of planning, budgeting, monitoring evaluation and follow-up managing competition. Role of ED in economic development of a country-Overview of Indian social, political systems and their implications for decision making by individual entrepreneurs- Economic system and its implications for decision making by individual entrepreneurs- Social responsibility of business. Morals and ethics in enterprise management- SWOT analysis- Government schemes and incentives for promotion of entrepreneurship.

Unit-V

Government policy on small and medium enterprises (SMEs)/SSIs/MSME sectors- Venture capital (VC), contract farming (CF) and joint ventures (JV), public-private partnerships (PPP)- Overview of agricultural engineering industry, characteristics of Indian farm machinery industry.

References:

1. Harsh, S.B., Conner, U.J. and Schwab, G.D. 1981.Management of the Farm Business. Prentice Hall Inc., New Jersey.
2. Joseph, L. Massie. 1995. Essentials of Management. Prentice Hall of India Pvt. Ltd., New Delhi.
3. Omri Rawlins, N. 1980. Introduction to Agribusiness. Prentice Hall Inc., New Jersey
4. Gittenger Price, J. 1989. Economic Analysis of Agricultural Projects. John Hopkins University, Press, London.
5. Thomas W Zimmer and Norman M Scarborough. 1996. Entrepreneurship. Prentice-Hall, New Jersey.
6. Mark J Dollinger. 1999. Entrepreneurship Strategies and Resources. Prentice-Hall, Upper Saddal Rover, New Jersey.
7. Khanka S S. 1999. Entrepreneurial Development. S. Chand and Co. New Delhi.
8. Mohanty S K. 2007. Fundamentals of Entrepreneurship. Prentice Hall India Ltd., New Delhi

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Semester – II

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Fluid Mechanics & Open Channel Hydraulics	BTA-5204	2-3 hr	3

Course Outcomes:

- **CO 1** Understand the properties of fluid
- **CO 2** Know the behaviour of the fluid at rest
- **CO 3** Know the behaviour of the fluid in motion
- **CO 4** Apply the principles of fluid mechanics to design simple fluid mechanical systems in engineering

Course Contents:

UNIT–I

Properties of fluids: Ideal and real fluid. Pressure and its measurement, pascal's law, pressure forces on plane and curved surfaces, centre of pressure, buoyancy, meta centre and meta centric height, condition of floatation and stability of submerged and floating bodies;

UNIT–II

Kinematics of fluid flow: Lagrangian and Eulerian description of fluid motion, continuity equation, path lines, streak lines and stream lines, stream function, velocity potential and flow net. Types of fluid flow, translation, rotation, circulation and vorticity, Vortex motion;

UNIT–III

Dynamics of fluid flow, Bernoulli's theorem, venturimeter, orifice meter and nozzle, siphon; Laminar flow: Stress strain relationships, flow between infinite parallel plates both plates fixed, one plate moving, discharge, average velocity; Laminar and turbulent flow in pipes, general equation for head loss Darcy, Equation, Moody's diagram, Minor and major hydraulic losses through pipes and fittings, flow through network of pipes, hydraulic gradient and energy gradient;

UNIT–IV

Flow through orifices (Measurement of Discharge, Measurement of Time),Flow through Mouthpieces, Flow over Notches , Flow over weirs, Chezy's formula for loss of head in pipes, Flow through simple and compound pipes, Open channel design and hydraulics: Chezy's formula, Bazin's formula, Kutter's Manning's formula,

UNIT–V

Velocity and Pressure profiles in open channels, Hydraulic jump; Dimensional analysis and similitude: Rayleigh's method and Buckingham's `Pi' theorem, types of similarities, dimensional analysis, dimensionless numbers. Introduction to fluid machinery.

References:

1. Bansal, R.K. A Text book of Fluid Mechanics, Laxmi Publications, New Delhi.
2. Ramanuthan, S. Hydraulics, Fluid Mechanics & Hydraulic Machines, Dhanpat rai & Sons, Delhi.
3. Khurmi, R.S. Hydraulics & Fluid Mechanics, S.Chand & Co.Ltd., New Delhi.
4. Modi, P.N. and Seth, S.M. Hydraulics & Fluid Mechanics, Standard Book House, Delhi
5. Paul, J. C. and Panigrahi, B. Practical Manual in Fluid Mechanics, CAET, OUAT, Bhubaneswar

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Semester – II

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Strength of Materials	BTA-5205	2-3 hr	2

Course Outcomes:

CO 1 Understand the importance of strength parameters

CO 2 Calculate deflection, bending moment and shear force to assess the stress conditions

CO 3 Know the techniques to calculate unknown forces in 2D structures.

Course Contents:

UNIT–I

Slope and deflection of beams using integration techniques, moment area theorems and conjugate beam method.

UNIT–II

Columns and Struts. Riveted and welded connections. Stability of masonry dams.

UNIT–III

Analysis of statically intermediate beams. Propped beams. Fixed and continuous beam analysis using superposition, three moment equation and moment distribution methods

References

1. Khurmi R.S. 2001. Strength of Materials S. Chand & Co., Ltd., New Delhi.
2. Junarkar S.B. 2001.Mechanics of Structures (Vo-I). Choratar Publishing House, Anand.
3. Ramamrutham S. 2003. Strengths of Materials. Dhanpat Rai and Sons, Nai Sarak, New Delhi.

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Semester – II

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Workshop Technology and Practice	BTA-5206	2-3 hr	3

Course Outcomes:

- **CO 1** Understand the importance of workshop technology.
- **CO 2** Analyze the different applications of different shops.
- **CO 3** Know the basic manufacturing process involved for production of different machine elements.

Course Contents:

UNIT–I

Introduction to various carpentry tools, materials, types of wood and their characteristics and Processes or operations in wood working;

UNIT–II

Introduction to Smithy tools and operations;

UNIT–III

Introduction to welding, types of welding, Oxyacetylene gas welding, types of flames, welding techniques and equipment. Principle of arc welding, equipment and tools. Casting processes;

UNIT–IV

Classification, constructional details of center lathe, Main accessories and attachments. Main operations and tools used on center lathes. Types of shapers, Constructional details of standard shaper. Work holding devices, shaper tools and main operations.

UNIT–V

Types of drilling machines. Constructional details of pillar types and radial drilling machines. Work holding and tool holding devices. Main operations. Twist drills, drill angles and sizes. Types and classification. Constructional details and principles of operation of column and knee type universal milling machines. Plain milling cutter. Main operations on milling machine.

References

1. Raghuwansi B S. 2009. A Course in Workshop Technology (Vol. I and II). Dhanpat Rai and Sons, 1682, Nai Sarak, New Delhi.
2. Hajra Choudhury S. K., Roy Nirjhar, Hajra ChoudhuryA. K. 2010. Elements of Workshop technology (Vol. I and II). Media Promoters and Publishers Pvt.Ltd., Mumbai.
3. Chapman W A J. 1989. Workshop Technology (Vol. I and II). Arnold Publishers (India) Pvt. Ltd., AB/9, Safdarjung Enclave, New Delhi.
4. Khurmi R.S. & Gupta J.K.2008. A Text Book of Workshop Technology. S.Chand & Company Ltd., New Delhi.

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Semester – II

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Theory of Machines	BTA-5207	2-3 hr	2

Course Outcomes:

CO 1 Understand the importance of theory behind the machine &structure.

CO 2 Analyse the relative motion between the various parts of machine, and forces which act on them.

CO 3 The knowledge of this subject is very essential for an engineer in designing the various parts of the machine.

CO 4 Recommend methods for balancing a component.

Course Contents:

UNIT–I

Elements, links, pairs, kinematics chain, and mechanisms. Classification of pairs and mechanisms. Lower and higher pairs. Four bar chain, slider crank chain and their inversions.

UNIT–II

Determination of velocity and acceleration using graphical (relative velocity and acceleration) method. Instantaneous centers.

UNIT–III

Types of gears. Law of gearing, velocity of sliding between two teeth in mesh.Involute and cycloidal profile for gear teeth. Spur gear, nomenclature, interference and undercutting. Introduction to helical, spiral, bevel and worm gear. Simple, compound, reverted, and epicyclic trains. Determining velocity ratio by tabular method. Turning moment diagrams, coefficient of fluctuation of speed and energy, weight of flywheel, flywheel applications.

UNIT–IV

Belt drives, types of drives, belt materials. Length of belt, power transmitted, velocity ratio, belt size for flat and V belts. Effect of centrifugal tension, creep and slip on power transmission, Chain drives.

UNIT–V

Types of friction, laws of dry friction. Friction of pivots and collars. Single disc, multiple disc, and cone clutches. Rolling friction, anti-friction bearings. Types of governors. Constructional details and analysis of Watt, Porter, Proell governors. Effect of friction, controlling force curves. Sensitiveness, stability, hunting, isochronism, power and effort of a governor. Static and dynamic balancing. Balancing of rotating masses in one and different planes.

References

1. Khurmi R.S. and Gupta J.K.2010. A Text Book of Theory of Machines. Euresia Publishing House(Pvt) Ltd., New Delhi.
2. Bansal R.K.2009. A Text Book of Theory of Machines. Laxmi Publications (P) Ltd., New Delhi.
3. Ballaney P.L. 2006. A Text Book of Theory of Machines. Khanna Publishers, New Delhi.
4. Ratan S.S. 2010. A Text Book of Theory of Machines. Tata McGraw Hill Publishing Company Ltd., New Delhi.

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Semester – II

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Web Designing and Internet Applications	BTA-5208	2-3 hr	2

Course Outcomes:

CO 1 Understand what is internet, the necessity of it.
CO 2 Know the web design principles
CO 3 Know using Java script how to make interactive Webpages.

Course Contents:

UNIT–I Basic principles in developing a web designing, Planning process, Five Golden rules of web designing, Designing navigation bar, Page design, Home Page Layout , Design Concept. Basics in Web Design,

UNIT–II Brief History of Internet, World Wide Web , creation of a web site, Web Standards , Audience requirement. .

UNIT–III Introduction to JavaScript, variables & functions, Working with alert, confirm and prompt, Connectivity of Web pages with databases; Project.

References:

1. Jennifer Niederst Robbins. Developing web design. latest edition.
2. Frain and Ben. Responsive Web Design with HTML5.
3. Nicholas c.Zakas. Java Script for Web Developers.
4. George Q. Huang, K. L Mak. Internet Applications in Product Design and Manufacturing. ISBN:3540434658.

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Semester – III

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Principles of Horticultural Crops and Plant Protection	BTA-5301	2-3 hr	2

Course Outcomes:

CO 1 Identification of different horticultural crops in the field
CO 2 Practice different methods of macro and micro-propagation
CO 3 Getting acquainted with different types of basic fertilizers, organic manures & bio-fertilizers.
CO 4 Understand when to apply fertilizer and irrigation water to horticultural crops
CO 5 Practice different training and pruning methods
CO 6 Recognize the maturity time for harvesting of crop
CO 7 Know the vegetable seed extraction technique
CO 8 Importance of nursery raising
CO 9 Basics of preparation of potting mixture, potting and repotting of plants
CO 10 Identification and use the garden tools
CO 11 Identify different major pests and diseases of horticultural crops

Course Contents:

UNIT-I

Scope of horticultural. Soil and climatic requirements for fruits, vegetables and floriculture crops, improved varieties.

UNIT-II

Criteria for site selection, layout and planting methods nursery raising, commercial varieties/hybrids, sowing and planting times and methods, seed rate and seed treatment for vegetable crops.

UNIT-III

Macro and micro propagation methods, plant growing structures pruning and training, crop coefficients, water requirements and critical stages.

UNIT-IV

Fertilizer application, fertigation, irrigation methods harvesting, grading and packaging, post harvest practices Garden tools, management of orchard, Extraction and storage of vegetables seeds.

UNIT-V Major pests and diseases and their management in horticulture crops.

References:

1. Bansal.P.C. 2008.Horticulture in India.CBS Publishers and Distributors, New Delhi.
2. Saraswathy, S., T.L.Preethi, S.Balasubramanyan, J. Suresh, N.Revathy and S.Natarajan. 2007.
3. Postharvest management of Horticultural Crops. Agrobios Publishers, Jodhpur.

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Semester – III

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Principles of Agronomy	BTA-5302	2-3 hr	3

Course Outcomes:

CO 1 Develop knowledge of the crop types, variety, seasons of sowing, sowing/transplanting methods & crop development stages.

CO 2 Understand the effect of weather parameters on crop production

CO 3 Develop knowledge on different tillage practices in the crop field

CO 4 Learn the time and methods of application of different types of Fertilizers & Organic Manures

CO 5 Distinguish between soil water constants and their role in crop production

CO 6 Distinguish weeds and know the methods of weed control

CO 7 Aware about various cropping patterns and their application

Course Contents:

UNIT-I

Introduction and scope of agronomy, Classification of crops, Effect of different weather parameters on crop growth and development

UNIT-II

Principles of tillage, tilth and its characteristics. Crop seasons. Methods, time and depth of sowing of major field crops.

UNIT-III

Methods and time of application of manures and fertilizers. Organic farming-Sustainable agriculture.

UNIT-IV

Soil water plant relationship, crop coefficients, water requirement of crops and critical stages for irrigation.

UNIT-V

Weeds and their control, crop rotation, cropping systems, Relay cropping and mixed cropping.

References:

1. Principles of Agronomy by Yellamanda Reddy T and Sankara Reddy G H, Kalyani Publishers
2. Principles and practices of Agronomy by S. S. Singh
3. Fundamentals of Agronomy by G. C. Dey
4. Principles of Agronomy by S.R. Reddy

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Semester – III

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Communication Skills and Personality Development	BTA-5303	2-3 hr	2

Course Outcomes:

CO 1 Develop academic and social English competencies in speaking,
CO 2 Listening and pronunciation, develop reading and writing skills
CO 3 Learn grammar and usage, vocabulary, syntax, and rhetorical patterns.

Course Content:

UNIT-I
Communication Skills: meaning and process of communication, types of communication: verbal and non-verbal communication

UNIT-II
Grammar: Structural and functional grammar

UNIT-III
Listening and note taking: writing skills: field diary and lab record; indexing, footnote and bibliographic procedures

UNIT-IV
Reading and comprehension of general and technical articles, précis Writing, summarizing, abstracting; individual and group presentations,

UNIT-V
Oral presentation impromptu presentation, public speaking; Group discussion. Organizing seminars and conferences.

References:

1. Raman, Meenakshi and Prakash Singh. 2000. Business Communication. Oxford University Press.
2. Kumar, Sanjay and Pushpa Lata. 2011. Communication Skills. Oxford University Press.
3. Thomson, A.J and A.V Martinet. 1977. A Practical Englsih Grammar. Oxford University Press.
4. Seely, John. Oxford Guide to Effective Writing and Speaking. Oxford University Press.

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Semester – III

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Engineering Mathematics III	BTA-5304	2-3 hr	3

Course Outcomes:

CO 1 Know about finite difference methods, operators and their relations

CO 2 Understand Newton's forward and backward interpolation

CO 3 Develop knowledge Beselles's and Stirling's central difference interpolations

CO 4 Know about Lagranges interpolation formula

CO 5 Know about numerical integration by trapezoidal and Simpson's rule

CO 6 Understand the rules for finding complimentary function and particular integral

CO 7 Apply Taylor's series, Euler's and modified Euler's method

CO 8 Know about properties of Laplace transform, inverse of Laplace transform and their applications

CO 9 Application of Laplace's transform to solve differential and simultaneous differential equations

Course Contents:

UNIT-I

Finite difference, various difference operators and their relationships. factorial notation, interpolation with equal integrals. Newton's forward and backward interpolation formula.

UNIT-II

Newton's divided difference formula. Lagrange's interpolation formula. numerical differentiations, numerical integrations.

UNIT-III

Difference equations and their solutions, numerical solutions of ordinary differential equations by Picard's Taylor's series. Fuller's and modified Fuller's methods. Runga-Kutta method.

UNIT-IV

Laplace transformation and its applications to the solutions of ordinary and simultaneous differential equations.

UNIT-V

Testing of Hypothesis-Level of Significance-Degrees of freedom-Statistical errors, Large sample test (Z-test), Small sample test t-test (One tailed, two tailed and Paired tests), Testing of Significance through variance (F-test), Chi -Square test, contingency table, Correlation, Regression.

References:

1. Chandel SRS.A Hand book of Agricultural Statistics. Achal Praskasam Masndir, Kanpur.
2. Agrawal B L. Basic Statistics. Wiley Eastern Ltd. New Age International Ltd.
3. Nageswara Rao G. Statistics for Agricultural Sciences.BS Publications.

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Semester – III

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Soil Mechanics	BTA-5305	2-3 hr	2

Course Outcomes:

CO 1 Know different engineering properties like moisture content, density, void ratio, porosity, grain size analysis etc.

CO 2 Know about different soil consistencies like Liquid limit, Plastic limit, and Shrinkage limit.

CO 3 Know about different stress conditions like shear stress, direct stress and stress due to different loading conditions

CO 4 Know about consolidation and compaction properties of soil

CO 5 Learn about different active and passive earth pressure on retaining walls

CO 6 Know about different stability conditions of slopes

Course Contents:

UNIT-I

Introduction of soil mechanics, field of soil mechanics, phase diagram, physical and index properties of soil, classification of soils,

UNIT-II

Effective and neutral stress, elementary concept of Boussinesq and Wester guards analysis, new mark influence chart. Seepage Analysis; Quick condition -two dimensional flow-Laplace equation, Velocity potential and stream function, Flow net construction. Shear strength, Mohr stress circle, theoretical relationship between principle stress circle, theoretical relationship between principal stress

UNIT-III

Mohr coulomb failure theory, effective stress principle. Determination of shear parameters by direct shear test, triangle test & vane shear test. Numerical exercise based on various types of tests. Compaction, composition of soils standard and modified protector test, abbot compaction and Jodhpur mini compaction test field compaction method and control.

UNIT-IV

Consolidation of soil: Consolidation of soils, one dimensional consolidation spring analogy, Terzaghi's theory, Laboratory consolidation test, calculation of void ratio and coefficient of volume change, Taylor's and Casagrande's method, determination of coefficient of consolidation. Earth pressure: plastic equilibrium in soils, active and passive states,

UNIT-V

Rankine's theory of earth pressure, active and passive earth pressure for cohesive soils, simple numerical exercises. Stability of slopes: introduction to stability analysis of infinite and finite slopes friction circle method, Taylor's stability number.

References:

1. Punmia B C, Jain A K and Jain A K. 2005.Soil Mechanics and Foundations.Laxmi Publications (P) Ltd. New Delhi.
2. Ranjan Gopal and Rao A S R. 1993.Basic and Applied Soil Mechanics. Welley Easters Ltd., New Delhi.
3. Singh Alam. 1994. Soil Engineering Vol. I. CBS Publishers and Distributions, Delhi.

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Semester – III

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Design of Structures	BTA-5306	2-3 hr	2

Course Outcomes:

CO 1 Understand the importance of structural design

CO 2 Calculate loads on a structure

CO 3 Design small and medium RCC and steel structures

Course Contains:

UNIT- I

Loads, use of BIS Codes and Design of connections

UNIT-II

Design of structural steel members in tension, compression and bending.

UNIT-III

Design of steel roof truss

UNIT-IV

Analysis and design of singly and doubly reinforced sections, Shear, Bond and Torsion

UNIT-V

Design of Flanged Beams, Slabs, Columns, Foundations, Retaining walls and Silos.

References:

1. Junarkar, S.B. 2001. Mechanics of Structures Vol.I Charotar Publishing Home, Anand.
2. Khurmi R. S. 2001. Strength of materials. S. Chand & Company Ltd., 7361, Ram Nagar, New Delhi – 110055.
3. Kumar Sushil 2003. Treasure of R.C.C. Design. R.K. Jain. 1705-A, Nai Sarak , Delhi-110006, P.B.1074.

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Semester – III

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Machine Design	BTA-5307	2-3 hr	2

Course Outcomes:

CO 1 Understand the designer's work concerned with adaption of existing design.

CO 2 To modify the existing design into new idea by adopting a new material or different method of manufacturing.

CO 3 May take up the work of new design.

Course Contains:

UNIT-1

Meaning of design, Phases of design, design considerations. Common engineering materials and their mechanical properties. Types of loads and stresses, theories of failure, factor of safety, selection of allowable stress. Stress concentration. Elementary fatigue and creep aspects.

UNIT-II

Cotter joints, knuckle joint and pinned joints, turnbuckle. Design of welded subjected to static loads.

UNIT-III

Design of threaded fasteners subjected to direct static loads, bolted joints loaded in shear and bolted joints subjected to eccentric loading.

UNIT-IV

Design of shafts under torsion and combined bending and torsion.Design of keys. Design of muff, sleeve, and rigid flange couplings.

UNIT-V

Design of helical and leaf springs. Design of flat belt and V-belt drives and pulleys.Design of gears. Design of screw motion mechanisms like screw jack, lead screw, etc.Selection of anti-friction bearings.

References:

1. Khurmi R S and Gupta J K. 2014. A Text Book of Machine Design. S. Chand & Company Ltd., New Delhi.
2. Jain R K. 2013. Machine Design. Khanna Publishers, 2-B Nath Market, Nai Sarak, New Delhi.
3. Sharma P.C.and Agarwal D.K. 2010.Machine Design. S. K. Kataria & Sons, New Delhi-110002.

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Semester – III

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Thermodynamics, Refrigeration and Air Conditioning	BTA-5308	2-3 hr	3

Course Outcomes:

- **CO 1** Understand the basics of thermodynamics.
- **CO 2** Understand the principle of various refrigeration cycles.
- **CO 3** Understand the principle of psychometric processes and air conditioning.
- **CO 4** Design of air conditioning system.

Course Contains:

UNIT-I

Thermodynamics properties, closed and open system, flow and non-flow processes, gas laws, laws of thermodynamics, internal energy. Application of first law in heating and expansion of gases in non-flow processes. First law applied to steady flow processes.

UNIT-II

Carnot cycle, Carnot theorem.Entropy, physical concept of entropy, change of entropy of gases in thermodynamics process.Otto, diesel and dual cycles.

UNIT-III

Principles of refrigeration, - units, terminology, production of low temperatures, air refrigerators working on reverse Carnot cycle and Bell Coleman cycle. Vapour refrigeration -mechanism, P-V,P-S,P-H diagrams, vapor compression cycles, dry and wet compression, super cooling and sub cooling. Vapour absorption refrigeration system.

UNIT-IV

Common refrigerants and their properties. Design calculations for refrigeration system. Cold storage plants. Thermodynamic properties of moist air, perfect gas relationship for approximate calculation, adiabatic saturation process, wet bulb temperature and its measurement, psychometric chart and its use, elementary psychometric process.

UNIT-V

Air conditioning – principles –Type and functions of air conditioning, physiological principles in air conditioning, air distribution and duct design methods, fundamentals of design of complete air conditioning systems – humidifiers and dehumidifiers – cooling load calculations, types of air conditioners – applications.

References:

1. Kothandaraman C P Khajuria P R and Arora S C. 1992.A Course in Thermodynamics and Heat Engines. Dhanpet Rai and Sons, 1682 Nai Sarak, New Delhi.
2. Khurmi R S. 1992. Engineering Thermodynamics. S Chand and Co. Ltd., Ram Nagar, New Delhi.
3. Mathur M L and Mehta F S. 1992.Thermodynamics and Heat Power Engineering.
4. Dhanpat Rai and Sons 1682 Nai Sarak, New Delhi.
5. Ballney P. L. 1994. Thermal Engineering. Khanna Publishers, New Delhi.
6. Nag P K.1995. Engineering Thermodynamics. Tata McGraw Hill Publishing Co. Ltd., New Delhi.

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Semester – III

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Electrical Machines and Power Utilization Conditioning	BTA-5309	2-3 hr	3

Course Outcomes:

CO 1 Different types of circuits and their applications.
CO 2 Principles and operation of transformers, DC machines and motors.
CO 3 Various methods of power measurement

Course Contains:

UNIT-I Magnetic circuit: Electro motive force & reluctance, Laws of magnetic circuits. Determination of ampere-turns for series magnetic circuits. Determination of ampere-turns for parallel magnetic circuits. B-H curve, hysteresis and eddy current losses.

UNIT-II Transformer: Principle of working, construction of single phase transformer, EMF equation. Phasor diagram on load, leakage reactance. Transformer on load, equivalent circuit.Voltage regulation, power and energy efficiency. Open circuit and short circuit tests.

UNIT-III D.C Machines: Principle operation and performance of dc machine (generator and motor). EMF and torque equations. Types of D.C. Machine. Armature reaction. Commutation. Excitation of dc generator and their characteristics. Dc motor characteristics. Starting of shunt and series motor and starters. Speed control D.C motor by using field and armature control method.

UNIT-IV Three phase induction motor: Construction, operation and equivalent circuit. Phasor diagram of poly-phase induction motor. Effect of rotor resistance and torque equation. Starting poly-phase induction motor. Speed control methods of poly-phase induction motor.

UNIT-V Single phase induction motor: Single phase induction motor: double field revolving theory. Equivalent circuit and characteristics. Phase split and shaded pole motors, Disadvantage of low power factor and power factor improvement. Various methods of single and three phase power measurement, Series and parallel resonance. Q-factor & bandwidth, line and phase quantity in star & delta network.

References:

1. Theraja B L & Theraja AK 2005. A text book of Electrical Technology. Vol. II S. Chand & Company LTD., New Delhi.
2. Thareja B L & Theraja AK. 2005. A text book of Electrical Technology. Vol.I S. Chand & Company LTD., New Delhi.
3. Vincent Del Toro. 2000. Electrical Engineering Fundamentals. Prentice-Hall of India Private LTD., New Delhi.
4. Anwani M L. 1997. Basic Electrical Engineering. Dhanpat Rai & Co. (P) LTD. New Delhi.

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Semester – IV

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Building Construction and Cost Estimation	BTA-5401	2-3 hr	2

Course Outcomes:

CO 1 Understand the importance of various building materials for construction work

CO 2 Know about various components of a building with items of work

CO 3 Know about methods of construction of agricultural buildings, sloped and flat roofed buildings

CO 4 Know about preparation of various types of estimates of buildings

CO 5 To take measurements, find depreciation, cost-in-use, benefit-to-cost, net benefits, payback period etc.

Course Contents:

UNIT-I

Building Materials: Rocks, Stones, Bricks Properties and varieties of Tiles, Lime, Cement, Concrete, Sand. Glass, Rubber, Plastics, iron, Steel, Aluminum, Copper, Nickel. Timber. **Building components**: Lintels, Arches, stair cases, Different types of floors, Finishing: Damp Proofing and water proofing, Plastering, pointing, white washing and distempering – Painting,

UNIT-II

Building design, Design procedures, Technology, building construction, Types of agricultural buildings and related needs, application of design theory and practice to the conservation, sloped and flat roof buildings,

UNIT-III

Construction economics: Preliminary estimates, Detailed Estimates of Buildings source of cost information, use of cost analyses for controlling design, Factors affecting building costs;

UNIT-IV

Cost evaluation of design and planning alternatives for building and estate development, Measurement and pricing,

UNIT-V

Economic methods for evaluating investments in buildings and building systems: cost-in-use, benefit-to-costs and savings-to-investment ratios, rate of return, net benefits, payback.

References:

1. Punmia, B.C., Jain, A.K. Building Construction, Laxmi Publications(P) ltd. New Delhi
2. Duggal, S.K.Building Materials, New Age International Publishers.
3. Sane, Y.S.Planning and Designing of Buildings
4. Rangwala, S.C.1994. Engineering Materials, Charotar Publishing House, Anand

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Semester – IV

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Auto CAD Applications	BTA-5402	2-3 hr	2

Course Outcomes:

CO 1 To introduce to the students the importance of machine drawing in engineering applications.

CO 2 To impart the knowledge of assembly drawing and production drawing of machine Components.

CO 3 The Computer Graphics course will enable the student to gain knowledge on how computers are integrated at various levels of planning and manufacturing.

CO 4 To understand the flexible manufacturing system and to handle the product data and various software used for manufacturing.

Course Contents:

Application of computers for design. CAD- Overview of CAD window – Explanation of various options on drawing screen. Study of draw and dimension tool bar. Practice on draw and dimension tool bar. Study of OSNAP, line thickness and format tool bar. Practice on OSNAP, line thickness and format tool bar. Practice on mirror, offset and array commands. Practice on trim, extend, chamfer and fillet commands. Practice on copy, move, scale and rotate commands. Drawing of 2 D- drawing using draw tool bar. Practice on creating boundary, region, hatch and gradient commands. Practice on Editing polyline- PEDIT and Explode commands. Setting of view ports for sketched drawings. Printing of selected view ports in various paper sizes. 2D- drawing of machine parts with all dimensions and allowances- Foot step bearing and knuckle joint. Sectioning of foot step bearing and stuffing box. Drawing of hexagonal, nut and bolt and other machine parts. Practice on 3-D commands-Extrusion and loft. Practice on 3-D commands-on sweep and press pull. Practice on 3-D Commands- revolving and joining. Demonstration on CNC machine and simple problems.

References:

1. Rao P.N. 2002. CAD/CAM Principles and Applications. McGraw-Hill Education Pvt. Ltd., New Delhi.
2. Sareen Kuldeep and Chandan Deep Grewal. 2010. CAD/CAM Theory and Practice. S. Chand & Company Ltd., New Delhi.
3. Zeid Ibrahim. 2011. Mastering CAD/CAM with Engineering. McGraw-Hill Education Pvt.
4. Ltd., New Delhi.

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Semester – IV

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Applied Electronics and Instrumentation	BTA-5403	2-3 hr	3

Course Outcomes:

- **CO 1** Know the operating principle of various electronics device.
- **CO 2** The use of different measuring instrument in different sector.
- **CO 3** Characteristics of diode and transistor.
- **CO 4** Application of OPAMP in various purposes.
- **CO 5** Application of diode as clipper and clamper circuit.
- **CO 6** Various types of transducers.

Course Contains:

Unit-I Semiconductors.p—n junction.V—I characteristics of p—n junction.diode as a circuit element.rectifier. clipper. damper, voltage multiplier, capacitive filter. diode circuits for OR, AND (both positive and negative logic)

UNIT-II Bipolar junction transistor: operating point. Classification(A.B & C) of amplifier.various biasing methods (fixed. self potential divider). h-parameter model of a transistor. analysis of small signal. CE amplifier.phase shift oscillator, analysis of differential amplifier using transistor.

UNIT-III Ideal OP-AMP characteristics. linear and non-linear applications of OP-AMP (adder. subtractor. integrator, active rectifier. comparator. differentiator. differential, instrumentation amplifier and oscillator).

UNIT-IV Zener diode voltage regulator. transistor series regulator. current limiting. OP-AMP voltage regulators.Bmap). binary ladder D/A converter, successive approximation A/D converter,

UNIT-V Generalized instrumentation, measurement of displacement. temperature. velocity, force and pressure using potentiometer. resistance thennometer. thermocouples. Bourclen tube.LVDT.strain gauge and tacho-generator.

References:

1. Mehta V. K. 2001.Principles of Electronics. S. Chand and Co., New Delhi.
2. Shaney A. K. 1997.Measurement of Electrical and Electronic Instrumentation. Khanna Publications
3. Chowdary Roy. 2008. Integrated Electronics. John Wiley International.
4. Kumar Anand. 2003. Digital Electronics. PHI. Publications.
5. Gupta Sanjeev, Gupta Santosh. 2014. Electronic Devices and Circuits. Dhanpat Rai Publications.

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New Scheme Based On ICAR Flexible Curriculum

Semester – IV

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Tractor and Automotive Engines	BTA-5404	2-3 hr	3

Course Outcomes:

CO 1 Understand the theory of engines, their types and usefulness

CO 2 Understand the working principle of different systems of internal combustion engines and tractor.

CO 3 To study the working principle, maintenance and adjustment of different engine systems

Course Contains:

UNIT-I

Study of **sources of farm power** –conventional & non-conventional energy sources. Classification of tractors and IC engines. Review of thermodynamic principles of IC (CI & SI) engines and deviation from ideal cycle. General energy equation and heat balance sheet. Study of mechanical, thermal and volumetric efficiencies. Study of engine components their construction, operating principles and functions. Study of engine strokes and comparison of 2-stroke and 4-stroke engine cycles and CI and SI engines.

UNIT-II

Study of Engine Valve systems, valve mechanism, Valve timing diagram, and valve clearance adjustment. Study of cam profile, valve lift and valve opening area. Study of importance of air cleaning system. Study of types of air cleaners and performance characteristics of various air cleaners. Study of fuel supply system. Study of fuels, properties of fuels, Calculation of air-fuel ratio. Study of tests on fuel for SI and CI engines.

UNIT-III

Study of fuel injection system – Injection pump, their types, working principles. Fuel injector nozzles – their types and working principle. Study of carburetion system, carburetors and their main functional components. Study of ignition system of SI engines. Study of detonation and knocking in IC engines. Engine governing – need of governors, governor types and governor characteristics.

UNIT-IV

Study of lubrication system – need, types, functional components. Study of lubricants – physical properties, additives and their application. Engine cooling system – need, cooling methods and main functional components.

UNIT-V

Study of need and type of thermostat valves. Additives in the coolant. Study of radiator efficiency. Study of electrical system including battery, starting motor, battery charging, cut-out, etc. Comparison of dynamo and alternator. Familiarization with the basics of engine testing.

References:

1. Liljedahl J B and Others.Tractors and Their Power Units.
2. Rodichev V and G Rodicheva.Tractors and Automobiles.
3. Mathur ML and RP Sharma.A course in Internal Combustion Engines.
4. Singh Kirpal. Automobile Engineering – Vol II.
5. Heitner Joseph. Automotive Mechanics : Principles and Practices.

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Semester – IV

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Engineering Properties of Agricultural Produce	BTA-5405	2-3 hr	2

Course Outcomes:

CO 1 Understand the various engineering properties like physical, thermal, frictional, aerodynamic, rheological and electrical properties of agricultural produce.

CO 2 Know the methods of measurement physical, thermal, frictional properties of granular Material

CO 3 Knowledge of Rheological and Electrical properties of Biological material

CO 4 Understand the importance of these properties in handling processing machines and storage structures

Course Contains:

UNIT-I

Classification and importance of engineering properties of Agricultural Produce, shape, size, roundness, sphericity, volume, density, porosity, specific gravity, surface area of grains, fruits and vegetables

UNIT-II

Thermal properties, Heat capacity, Specific heat, Thermal conductivity, Thermal diffusivity, Heat of respiration; Co-efficient of thermal expansion.

UNIT-III

Friction in agricultural materials; Static friction, Kinetic friction, rolling resistance, angle of internal friction, angle of repose, Flow of bulk granular materials. Aero dynamics of agricultural products, drag coefficients, terminal velocity.

UNIT-IV

Rheological properties; force, deformation, stress, strain, elastic, plastic and viscous behaviour, Newtonian and Non-Newtonian liquid, Visco-elasticity, Newtonian and Non-Newtonian fluid, Pseudo-plastic, Dilatant, Thixotropic, Rheopectic and Bingham Plastic Foods, Flow curves.

UNIT-V

Electrical properties; dielectric loss factor, loss tangent, A.C. conductivity and dielectric constant, method of determination. Application of engineering properties in handling processing machines and storage structures.

References:

1. Singhal OP and Samuel DVK. 2003. Engineering Properties of Biological Materials. Saroj Prakasan.NewDelhi
2. Mohesin, N.N. 1980. Physical Properties of Plants & Animals. Gordon & Breach Science Publishers, New York.
3. Rao, M.A. and Rizvi,S.H., 1995. Engineering Properties of Foods. Marcel Dekker Inc. New York.
4. Stroshine, R. 1998. Physical Properties of Agricultural Materials and Food Products. Course Manual. Purdue University. USA.
5. Serpil S and Servet G S.2005. Physical Properties of Foods. (Springer Science Business Media, LLC, 233 Spring Street, New York

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Semester – IV

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Watershed Hydrology	BTA-5406	2-3 hr	2

Course Outcomes:

- **CO 1** Understand the relevance of various components of hydrologic cycle, which are responsible for spatial and temporal distribution of water availability in any region.
- **CO 2** Quantify different hydrological processes, know their methods of analysis for application in hydrological studies.
- **CO 3** Apply ideas and techniques in related areas of watershed development, water harvesting, minor irrigation, drought and flood control etc.

Course Contents:

UNIT-I

Hydrologic cycle, precipitation and its forms, rainfall measurement and estimation of mean rainfall, frequency analysis of point rainfall.

UNIT-II

Mass curve, hyetograph, depth-area-duration curves and intensity-duration-frequency relationship. **Hydrologic processes**-Interception, infiltration -factors influencing, measurement and indices. Evaporation estimation and measurement.

UNIT-III

Runoff - Factors affecting, measurement, stage - discharge rating curve, estimation of peak runoff rate and volume, Rational method, Cook's method and SCS curve number method.

UNIT-IV

Geomorphology of watersheds – Linear, aerial and relief aspects of watersheds- stream order, drainage density and stream frequency. Hydrograph - Components, base flow separation, unit hydrograph theory, S-curve, synthetic hydrograph, applications and limitations.

UNIT-V

Stream gauging - discharge rating curves, flood peak, design flood and computation of probable flood. Flood routing – channel and reservoir routing. Drought – classification causes and impacts, drought management strategy.

References:

1. Chow, V.T., D.R. Maidment and L.W. Mays. 2010. Applied Hydrology, McGraw Hill Publishing Co., New York.
2. Jaya Rami Reddy, P. 2011.A Text Book of Hydrology. University Science Press, New Delhi.
3. Linsley, R.K., M.A. Kohler, and J.L.H. Paulhus. 1984. Hydrology for Engineers. McGraw-Hill Publishing Co., Japan.
4. Mutreja, K.N. 1990. Applied Hydrology. Tata McGraw-Hill Publishing Co., New Delhi.
5. Raghunath, H.M. 2006. Hydrology: Principles Analysis and Design. Revised 2nd Edition, New
6. Subramanya, K. 2008. Engineering Hydrology.3rd Edition, Tata McGraw-Hill Publishing Co., New Delhi.
7. Suresh, R. 2005. Watershed Hydrology.Standard Publishers Distributors, Delhi.
8. Varshney, R.S. 1986. Engineering Hydrology. Nem Chand and Brothers, Roorkee, U.P.

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Semester – IV

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Irrigation Engineering	BTA-5407	2-3 hr	3

Course Outcomes:

CO 1 Knowledge of the different methods to measure the soil moisture content in the field.

CO 2 Knowledge of various procedures to measure discharge of water flowing in the irrigation channel.

CO 3 Understand the different methodologies to compute the water requirement of crops.

CO 4 Know how to design the underground water conveyance system and irrigation channels

CO 5 Knowledge of different lining materials used to reduce the seepage loss in channels

CO 6 Knowledge of advance and recession in border and furrow irrigation, cut-off stream in furrow irrigation and irrigation efficiency etc.

CO 7 Measurement the bulk density, field capacity and wilting point

CO 8 Evaluate the border , the furrow and the check basin irrigation methods

CO 9 Be acquainted with accurate irrigation scheduling with proper methods of irrigation for different crops.

Course Contains:

UNIT-I

Major and medium irrigation schemes of India, purpose of irrigation, environmental impact of irrigation projects, source of irrigation water, present status of development and utilization of different water resources of the country

UNIT-II

Measurement of irrigation water: weir, flumes and orifices and other methods; open channel water conveyance system

UNIT-III

Design and lining of irrigation field channels, on farm structures for water conveyance, control & distribution; underground pipe conveyance system: components and design; land grading: criteria for land leveling, land leveling design methods,

UNIT-IV

Estimation of earth work; soil water plant relationship: soil properties influencing irrigation management, soil water movement, infiltration, soil water potential, soil moisture characteristics, soil moisture constants, measurement of soil moisture, moisture stress and plant response;

UNIT-V

Water requirement of crops: concept of evapotranspiration (ET), measurement and estimation of ET, water and irrigation requirement of crops, depth of irrigation, frequency of irrigation, irrigation efficiencies; surface methods of water application: border, check basin and furrow irrigation-adaptability, specification and design considerations.

References:

1. Michael A.M. 2012. Irrigation: Theory and Practice. Vikas Publishing House New Delhi
2. Majumdar D. K. 2013. Irrigation Water Management Principles. PHI learning Private Limited New Delhi 2nd Edition
3. Panigrahi, B. 2013. A Handbook on Irrigation and Drainage. New India Publishing Agency, New Delhi
4. Allen R. G., L. S. Pereira, D. Raes, M. Smith. 1998. Crop Evapotranspiration guidelines for computing crop water requirement. Irrigation and drainage paper 56, FAO of United Nations, Rome
5. Murthy VVN. 2013. Land and water Management Engineering. Kalyani Publishers, New Delhi.
6. Israelsen O W. and Hansen V. E and Stringham G. E. 1980. rrigation Principles and Practices, John Wiley & Sons, Inc. USA.

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Semester – IV

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Sprinkler and Micro irrigation Systems	BTA-5408	2-3 hr	2

Course Outcomes:

- **CO 1** To understand the importance of micro-irrigation (drip and sprinkler, role of Government for promotion,
- **CO 2** To acquaint the students about the components of micro irrigation systems, their design and lay out for efficient water, fertilizer and pesticides application
- **CO 3** To gain knowledge about the hydraulics of sprinkler and drip irrigation
- **CO 4** Benefit – cost analysis in micro irrigation systems

Course Contains:

UNIT-I

Sprinkler irrigation: adaptability, problems and prospects, types of sprinkler irrigation systems; design of sprinkler irrigation system

UNIT-II

Layout selection, hydraulic design of lateral, sub-main and main pipe line, design steps; selection of pump and power unit for sprinkler irrigation system; performance evaluation of sprinkler irrigation system: uniformity coefficient and pattern efficiency;

UNIT-III

Micro Irrigation Systems: types-drip, spray, & bubbler systems, merits and demerits, different components; Design of drip irrigation system: general considerations, wetting patters, irrigation requirement, emitter selection, hydraulics of drip irrigation system,

UNIT-IV

Design steps; necessary steps for proper operation of a drip irrigation system; maintenance of micro irrigation system: clogging problems, filter cleaning, flushing and chemical treatment;

UNIT-V

Fertigation: advantages and limitations of fertigation, fertilizers solubility and their compatibility, precautions for successful fertigation system, fertigation frequency, duration and injection rate, methods of fertigation.

References:

1. Keller Jack and Bliesner Ron D. 2001. Sprinkle and Trickle Irrigation. Springer Science business Media, New York .
2. Mane M.S. and Ayare B.L.2007. Principles of Sprinkler Irrigation systems, Jain Brothers, New Delhi.
3. Mane M.S and Ayare B.L. and Magar S.S.2006. Principles of Drip Irrigation systems, Jain Brothers, New Delhi.
4. Michael AM, Shrimohan and KR Swaminathan. Design and evaluation of irrigation methods, (IARI Monograph No.1). Water Technology Centre, IARI New Delhi.
5. Michael A.M. 2012. Irrigation: Theory and Practice. Vikas Publishing Vikas Pub. House New Delhi.
6. Choudhary M.L and Kadam U.S 2006. Micro irrigation for cash crops Westville Publishing House

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Semester – IV

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Fundamentals of Renewable Energy Sources	BTA-5409	2-3 hr	3

Course Outcomes:

CO 1 To harness the environment friendly RE sources and to enhance their contribution to the socio economic development.

CO 2 To meet and supplement rural energy needs through sustainable RE projects.

CO 3 To provide decentralized energy supply to agriculture, industry, commercial and household sector.

CO 4 On successful completion of the teaching program, the students should be able to evaluate, and select appropriate energy technologies to meet a given energy demand.

Course Contains:

UNIT-I

Concept and limitation of Renewable Energy Sources (RES), Criteria for assessing the potential of RES, Classification of RES, Solar, Wind, Geothermal, Biomass, Ocean energy sources, Comparison of renewable energy sources with non-renewable sources.

UNIT-II

Solar Energy: Energy available from Sun, Solar radiation data, solar energy conversion into heat through, Flat plate and Concentrating collectors, different solar thermal devices, Principle of natural and forced convection drying system, Solar Photo voltaic: p-n junctions. Solar cells, PV systems, Stand alone and grid connected solar power station, Calculation of energy through photovoltaic power generation and cost economics.

UNIT-III

Wind Energy: Energy available from wind, General formula, Lift and drag. Basis of Wind energy conversion, Effect of density, Frequency variances, Angle of attack, Wind speed, Types of Windmill rotors, Determination of torque coefficient, Induction type generators, working principle of wind power plant.

UNIT-IV

Bio-energy: Pyrolysis of Biomass to produce solid, liquid and gaseous fuels. Biomass gasification, Types of gasifier, various types of biomass cook stoves for rural energy needs.

UNIT-V

Biogas: types of biogas plants, biogas generation, factors affecting biogas generation and usages, design consideration, advantages and disadvantages of biogas spent slurry.

References:

1. Rai, G.D. 2013. Non-Conventional Energy Sources, Khanna Publishers, Delhi.
2. Rai, G.D., Solar Energy Utilization, Khanna Publishers, Delhi.
3. Khandelwal, K.C. & S. S. Mahdi. 1990. Biogas Technology- A Practical Handbook.
4. Rathore N. S., Kurchania A. K., Panwar N. L. 2007. Non-Conventional Energy Sources, Himanshu Publications.
5. Tiwari, G.N. and Ghoshal, M.K. 2005. Renewable Energy Resources: Basic Principles and Applications. Narosa Pub. House. Delhi.
6. Rathore N. S., Kurchania A. K., Panwar N. L. 2007. Renewable Energy, Theory and Practice, Himanshu Publications.

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Semester – V

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Tractor Systems and Controls	BTA-5501	2-3 hr	3

Course Outcomes:

CO 1 Importance of different systems in a Tractor
CO 2 Transmission systems including final drive
CO 3 Brake System
CO 4 Steering System
CO 5 Hydraulic system in tractor
CO 6 Power outlets
CO 7 Tractor Chassis mechanics
CO 8 Safety considerations in Tractor
CO 9 Standards of Tractor Testing

Course Contents:

UNIT-I

INTRODUCTION: Study of need for transmission system in a tractor. Transmission system – types, major functional systems. Study of clutch – need, types, functional requirements, construction and principle of operation. Familiarization with single plate, multi-plate, centrifugal and dual clutch systems.

UNIT-II

GEAR BOX: Study of Gear Box – Gearing theory, principle of operation, gear box types, functional requirements, and calculation for speed ratio. Study of differential system – need, functional components, construction, calculation for speed reduction. Study of need for a final drive.

UNIT-III

BRAKE, STEERING AND HYDRAULIC SYSTEM: Study of Brake system – types, principle of operation, construction, calculation for braking torque. Study of steering system – requirements, steering geometry characteristics, functional components, calculation for turning radius. Familiarization with Ackerman steering. Steering systems in track type tractors. Study of Hydraulic system in a tractor – Principle of operation, types, main functional components, functional requirements. Familiarization with the Hydraulic system adjustments and ADDC. Study of tractor power outlets – PTO.PTO standards, types and functional requirements.

UNIT-IV

TRACTION, WHEELS AND TYRES: Introduction to traction. Traction terminology. Theoretical calculation of shear force and rolling resistance on traction device. Study of wheels and tyres – Solid tyres and pneumatic tyres, tyre construction and tyre specifications. Study of traction aids. Study of tractor mechanics – forces acting on the tractor. Determination of CG of a tractor.

UNIT-V

DRAWBAR PULLAND TRACTOR TESTING: Determination and importance of moment of inertia of a tractor. Study of tractor static equilibrium, tractor stability especially at turns. Determination of maximum drawbar pull. Familiarization with tractor as a spring-mass system. Ergonomic considerations and operational safety. Introduction to tractor testing. Deciphering the engine test codes.

REFERENCES:

1. Liljedahl J B and Others.Tractors and Their Power Units.
2. Rodichev V and G Rodicheva.Tractors and Automobiles.
3. Singh Kirpal. Automobile Engineering – Vol I.
4. Heitner Joseph. Automotive Mechanics: Principles and Practices.
5. C.B.Richey. Agricultural Engineering Handbook.
6. John Deere. Fundamentals of Service Hydraulics.
7. Relevant BIS Test Codes for Tractors.

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Semester – V

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Farm Machinery and Equipment-I	BTA-5502	2-3 hr	3

Course Outcomes:

CO 1 Know about the farm machineries used in agricultural production.
CO 2 Know about construction, operation of different machines.
CO 3 Know about the operating parameters and performance of the machines.
CO 4 Know about the cost of operations and economics of the machines.
CO 5 To solve numerical on different chapters.
CO 6 Study the different components of the machines, their constructions.
CO 7 Materials of construction.
CO 8 Adjustments of different components to enhance performance.

Course Contents:

UNIT-I
INTRODUCTION: Introduction to farm mechanization. Classification of farm machines. Unit operations in crop production. Identification and selection of machines for various operations on the farm. Hitching systems and controls of farm machinery.

UNIT-II
EARTH MOVING EQUIPMENT: Calculation of field capacities and field efficiency. Calculations for economics of machinery usage, Comparison of ownership with hiring of machines. Introduction to seed -bed preparation and its classification. Familiarization with land reclamation and earth moving equipment. Introduction to machines used for primary tillage, secondary tillage, rotary tillage, deep tillage and minimum tillage. Measurement of draft of tillage tools and calculations for power requirement for the tillage machines.

UNIT-III
TILLAGE MACHINES: Introduction to tillage machines like mould -board plough, disc plough, chisel plough, sub-soiler, harrows, cultivators, Identification of major functional components. Attachments with tillage machinery. Introduction to sowing, planting & transplanting equipment. Introduction to seed drills, no-till drills, and strip-till drills.

UNIT-IV
STUDY OF SEED-DRILLS, PLANTERS AND FURROW OPENERS: Introduction to planters, bed-planters and other planting equipment. Study of types of furrow openers and metering systems in

drills and planters. Calibration of seed-drills/ planters. Adjustments during operation. Introduction to materials used in construction of farm machines.

UNIT-V

HEAT TREATMENT PROCESSES: Heat treatment processes and their requirement in farm machines. Properties of materials used for critical and functional components of agricultural machines. Introduction to steels and alloys for agricultural application. Identification of heat treatment processes specially for the agricultural machinery components.

REFERENCES:

1. Kepner RA, Roy Barger & EL Barger. Principles of Farm Machinery.
2. Smith HP and LH Wilkey.Farm Machinery and Equipment.
3. Culpin Claude. Farm Machinery.
4. Srivastava AC. Elements of Farm Machinery.
5. Lal Radhey and AC Datta.Agricultural Engineering.

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Semester – V

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Agricultural Structures and Environmental Control	BTA-5503	2-3 hr	3

Course Outcomes:

CO 1 Understand the importance of planning and lay out of a farmstead

CO 2 Know about various standards for various dairy, piggery, poultry and other farm structures.

CO 3 Know about the different farm storage structures, silos, compost pit, implement sheds, farm houses, threshing floors, farm roads, fencing, water supply, sewage systems, and septic tanks

CO 4 Know about rural electrification, concepts of eco system, bio-diversity, environmental pollution and control, solid waste, plant waste management

CO 5 To prepare estimate for different farm buildings, structures, roads, fencing and construction, repair and maintenance of farm structures.

Course Contents:

UNIT-I

Introduction: Planning and layout of farmstead. Scope, importance and need for environmental control, physiological reaction of livestock environmental factors, environmental control systems and their design, control of temperature, humidity and other air constituents by ventilation and other methods.

UNIT-II

Dairy& poultry farm structures: Livestock production facilities, BIS Standards for dairy, piggery, poultry and other farm structures. Design, construction and cost estimation of farm structures; animal shelters, compost pit, fodder silo, fencing and implement sheds, barn for cows, buffalo, poultry, etc.

UNIT-III

Storage of grains structures: Causes of spoilage, Water activity for low and high moisture food and its limits for storage, Moisture and temperature changes in grain bins; Traditional storage structures and their improvements, Improved storage structures (CAP, hermetic storage, Pusa bin, RCC ring bins), Design consideration for grain storage godowns, Bag storage structures, Shallow and Deep bin, Calculation of pressure in bins, Storage of seeds.

UNIT-IV

Water supply: Rural living and development, rural roads, their construction cost and repair and maintenance. Sources of water supply, norms of water supply for human being and animals, drinking water standards and water treatment suitable to rural community.

UNIT-V

Sewage system and septic tank: Site and orientation of building in regard to sanitation, community sanitation system; sewage system and its design, cost and maintenance, design of septic tank for small family. Estimation of domestic power requirement, source of power supply and electrification of rural housing.

References:

1. Pandey, P.H. Principles and practices of Agricultural Structures and Environmental Control, Kalyani Publishers, Ludhiana.
2. Ojha, T.P and Michael, A.M. Principles of Agricultural Engineering, Vol. I, Jain Brothers, Karol Bag, New Delhi.
3. Nathonson, J.A. Basic Environmental Technology, Prentice Hall of India, New Delhi. Venugopal Rao, P. Text Book of Environmental Engineering, Prentice Hall of India, New Delhi.
4. Garg, S.K. Water Supply Engineering, Khanna Publishers, New Delhi-6.
5. Dutta, B.N. Estimating and Costing in Civil Engineering, Duttta & CO, Lucknow.
6. Khanna, P.N. Indian Practical Civil Engineer's Hand Book, Engineer's Publishers, New Delhi.Sahay, K.M. and Singh, K.K. Unit Operations of Agricultural Processing, Vikas publishing pvt.Ltd, Noida.
7. Banerjee, G.C. A Text Book of Animal Husbandry, Oxford IBH Publishing Co, New Delhi.

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Semester – V

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Post Harvest Engineering of Cereals, Pulses & Oil Seeds	BTA-5504	2-3 hr	3

Course Outcomes:

CO 1 Know the different unit operations in processing of major cereals, pulses and oilseeds of the country and state

CO 2 Understand the principles behind the working principles of different machineries used for processing of cereals, pulses and oilseeds

CO 3 the basics of selection of appropriate machines/ equipment for various applications

CO 4 Know the different uses of byproducts obtained from cereals, pulses and oilseeds Processing.

Course Contents:

UNIT-I

Introduction: Cleaning and grading, aspiration, scalping; size separators, screens, sieve analysis, capacity and effectiveness of screens. Various types of separators: specific gravity, magnetic, disc, spiral, pneumatic, inclined draper, velvet roll, colour sorters, cyclone, shape graders.

UNIT-II

Size reduction machinery and law: Size reduction: principle, Bond's law, Kick's law, Rittinger's law, procedure (crushing, impact, cutting and shearing), Size reduction machinery: Jaw crusher, Hammer mill, Plate mill, Ball mill.Material handling equipment. Types of conveyors: Belt, roller, chain and screw. Elevators: bucket, Cranes & hoists. Trucks (refrigerated/ unrefrigerated), Pneumatic conveying.

UNIT-III

Dryer: moisture content and water activity; Free, bound and equilibrium moisture content, isotherm, hysteresis effect, EMC determination, Psychrometric chart and its use in drying, Drying principles and theory, Thin layer and deep bed drying analysis, Falling rate and constant rate drying periods, maximum and decreasing drying rate period, drying equations, Mass and energy balance, Shedd's equation, Dryer performance, Different methods of drying, batch-continuous; mixing-non-mixing, Sun-mechanical, conduction, convection, radiation, superheated steam, tempering during drying, Different types of grain dryers: bin, flat bed, LSU, columnar, RPEC, fluidized, rotary and tray. Mixing: Theory of mixing of solids and pastes, Mixing index, types of mixers for solids, liquid foods and pastes.

UNIT-IV

Milling: Milling of rice: Conditioning and parboiling, advantages and disadvantages, traditional methods, CFTRI and Jadavpur methods, Pressure parboiling method, Types of rice mills, Modern rice milling, different unit operations and equipment. Milling of wheat, unit operations and equipment. Milling of pulses: traditional milling methods, commercial methods, pre-conditioning, dry milling and wet milling methods.

UNIT-V

CFTRI and pantnagar methods. Pulse milling machines, Milling of corn and its products. Dry and wet milling. Milling of oilseeds: mechanical expression, screw press, hydraulic press, solvent extraction methods, preconditioning of oilseeds, refining of oil, stabilization of rice bran., Extrusion cooking: principle, factors affecting, single and twin screw extruders. By-products utilization.

References:

1. Chakraverty, A. Post Harvest Technology of cereals, pulses and oilseeds. Oxford & IBH publishing Co. Ltd., New Delhi.
2. Dash, S.K., Bebartta, J.P. and Kar, A. Rice Processing and Allied Operations. Kalyani Publishers, New Delhi.
3. Sahay, K.M. and Singh, K.K. 1994. Unit operations of Agricultural Processing. Vikas Publishing house Pvt. Ltd. New Delhi.
4. Geankoplis C. J. Transport processes and unit operations, Prentice Hall of India Pvt Ltd, New Delhi
5. Earle, R.L. 2003. Unit Operations in Food Processing. Pergamon Press. Oxford. U.K. Henderson, S.M., and Perry, R. L. Agricultural Process Engineering, Chapman and hall, London

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Semester – V

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Soil and Water Conservation Engineering	BTA-5505	2-3 hr	3

Course Outcomes:

CO 1 Understand water and wind erosion and their mechanisms.

CO 2 Know various agronomical and mechanical measures for controlling soil erosion and moisture conservation.

CO 3 Develop analytical thinking and problem solving skills in soil and water conservation engineering problems.

CO 4 Measure and estimate soil loss and sedimentation using different techniques.

CO 5 Design bunds, terraces, grassed waterways, wind breaks and shelter belts etc. and recommend them for better soil and moisture conservation in a watershed.

Course Contents:

UNIT-I

Erosion: Soil erosion - Introduction, causes and types - geological and accelerated erosion, agents, factors affecting and effects of erosion. Water erosion - Mechanics and forms - splash, sheet, rill, gully, ravine and stream bank erosion.

UNIT-II

Gullies and soil loss estimation: Gullies - Classification, stages of development. Soil loss estimation – Universal soil loss equation (USLE) and modified USLE. Rainfall erosivity - estimation by KE>25 and EI 30 methods.

UNIT-III

Soil erosion and agronomical measures: Soil erodibility– Topography, crop management and conservation practice factors. Measurement of soil erosion - Runoff plots, soil samplers. Water erosion control measures - agronomical measures - contour farming, strip cropping, conservation tillage and mulching.

UNIT-IV

Engineering measures: Bunds and terraces. Bunds - contour and graded bunds - design and surplus sing arrangements. Terraces - level and graded broad base terraces, bench terraces - planning, design and layout procedure, contour stonewall and trenching. Gully and ravine reclamation - principles of gully control - vegetative measures, temporary structures and diversion drains. Grassed waterways and design.

UNIT-V

Wind erosion: Factors affecting, mechanics, soil loss estimation and control measures - vegetative, mechanical measures, wind breaks and shelter belts and stabilization of sand dunes. Land capability classification. Rate of sedimentation, silt monitoring and storage loss in tanks.

References:

1. Singh Gurmel, C. Venkataraman, G. Sastry and B.P. Joshi. 1996. Manual of Soil and Water Conservation Practices. Oxford and IBH Publishing Co. Pvt. Ltd., New Delhi.
2. Mahnot, S.C. 2014. Soil and Water Conservation and Watershed Management. International Books and Periodicals Supply Service, New Delhi.
3. Mal, B.C. 2014. Introduction to Soil and Water Conservation Engineering. 2014. Kalyani Publishers.
4. Michael, A.M. and T.P. Ojha. 2003. Principles of Agricultural Engineering. Volume II. 4th Edition, Jain Brothers, New Delhi.
5. Murthy, V.V.N. 2002. Land and Water Management Engineering. 4th Edition, Kalyani Publishers, New Delhi.
6. Norman Hudson. 1985. Soil Conservation. Cornell University Press, Ithaka, New York, USA. Frevert, R.K., G.O. Schwab, T.W. Edminster and K.K. Barnes. 2009. Soil and Water Conservation Engineering, 4th Edition, John Wiley and Sons, New York.
7. Suresh, R. 2014. Soil and Water Conservation Engineering. Standard Publisher Distributors, New Delhi.

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Semester – V

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Watershed Planning and Management	BTA-5506	2-3 hr	2

Course Outcomes:

CO 1 Know the problems of watershed and characteristics and how to delineate a watershed.
CO 2 Different concept and factors affecting on watershed.
CO 3 Rainwater conservation technologies
CO 4 Monitoring and evaluation of a watershed along with PRA study and cost analysis
CO 5 How to delineate a watershed.

Course Contents:

UNIT-I
Introduction: Watershed- introduction and characteristics. Watershed development - problems and prospects, investigation, topographical survey, soil characteristics, vegetative cover, present land use practices and socio-economic factors.

UNIT-II
Watershed management - concept, objectives, factors affecting, watershed planning based on land capability classes, hydrologic data for watershed planning, delineation and prioritization of watersheds – sediment yield index.

UNIT-III
Water budgeting & Dry farming techniques: Water budgeting in a watershed. Management measures - rainwater conservation technologies - in-situ and ex-situ storage, water harvesting and recycling. Dry farming techniques - inter-terrace and inter -bund land management.

UNIT-IV
Components of watershed management: Integrated watershed management - concept, components, arable lands - agriculture and horticulture, non-arable lands - forestry, fishery and animal husbandry. Effect of cropping systems, land management and cultural practices on watershed hydrology.

UNIT-V
User groups pra and self-help groups: Watershed programme - execution, follow-up practices, maintenance, monitoring and evaluation. Participatory watershed management - role of watershed

associations, user groups and self-help groups. Planning and formulation of project proposal for watershed management programme including cost-benefit analysis.

References:

1. Ghanshyam Das. 2008. Hydrology and Soil Conservation Engineering: Including Watershed Management. 2nd Edition, Prentice-Hall of India Learning Pvt. Ltd., New Delhi.
2. Katyal, J.C., R.P. Singh, Shriniwas Sharma, S.K. Das, M.V. Padmanabhan and P.K. Mishra.1995. Field Manual on Watershed Management. CRIDA, Hyderabad.
3. Mahnot, S.C. 2014. Soil and Water Conservation and Watershed Management. International Books and Periodicals Supply Service. New Delhi.
4. Sharda, V.N., A.K. Sikka and G.P. Juyal. 2006. Participatory Integrated Watershed Management: A Field Manual. Central Soil and Water Conservation Research and Training Institute, Dehradun.
5. Singh, G.D. and T.C. Poonia. 2003. Fundamentals of Watershed Management Technology.

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Semester – V

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Drainage Engineering	BTA-5507	2-3 hr	2

Course Outcomes:

CO 1 Understand the drainage problems and their effect on crop production

CO 2 Know about the salt affected soils and their reclamation methods

CO 3 Distinguish between surface, sub-surface and non-conventional drainage methods

CO 4 Investigation methods of drainage problems

CO 5 Design drain dimensions and the spacing between drains

CO 6 Identify the drainage materials and study their strength prior to installation

CO 7 Evolve methods to design the surface and sub-surface drainage systems

Course Contents:

UNIT-I

Introduction: Water logging- causes and impacts; drainage, objectives of drainage, familiarization with the drainage problems of the state; surface drainage coefficient,

UNIT-II

Surface drainage: Design of surface drains; sub-surface drainage: purpose and benefits, investigations of design parameters-hydraulic conductivity, drainable porosity, water table;

UNIT-III

Sub-surface drainage: Derivation of Hooghoudt's and Ernst's drain spacing equations; design of subsurface drainage system; drainage materials, drainage pipes, drain envelope

UNIT-IV

Drainage installation: Layout, construction and installation of drains; drainage structures; vertical drainage; bio-drainage; mole drains; salt balance,

UNIT-V

Reclamation of saline & alkaline soils: leaching requirements, conjunctive use of fresh & saline water.

References:

1. Bhattacharya AK and Michael AM. 2013. Land Drainage, Principles , Methods and Applications. Vikas Publication House, Noida (UP).
2. Ritzema H.P.1994 Drainage Principles and Applications, ILRI Publication 16, Second Edition (Completely Revised).
3. Michael AM. and Ojha TP. 2014. Principles of Agricultural Engineering Vol-II 5th Edition.

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Semester – V

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Renewable Power Sources	BTA-5508	2-3 hr	3

Course Outcomes:

CO 1 To study the different energy consumption pattern and replace of renewable energy

CO 2 Design of community type and institutional type of biogas plants and its use for IC engine operation for power generation,

CO 3 Solar thermal and photovoltaic Power generation

CO 4 Detail design of wind mill and its power operation

CO 5 Ocean power generation through OTEC, wave and tide

CO 6 Power generation from urban waste and biomass gasification

CO 7 Design of mini and small hydro project

CO 8 Detail study of fuel cell and their application

CO 9 Use of animal energy for different agricultural operation.

Course Contents:

UNIT-I

Introduction: Energy consumption pattern & energy resources in India. Renewable energy options, potential and utilization.

UNIT-II

Biogas: Biogas technology and mechanisms, generation of power from biogas, Power generation from urban, municipal and industrial waste.

UNIT-III

Solar thermal and photovoltaic systems: Design & use of different commercial sized biogas plant. Solar thermal and photovoltaic Systems for power generation. Central receiver (Chimney) and distributed type solar power plant, OTEC, MHD, hydrogen and fuel cell technology.

UNIT-IV

Wind farms: Aero-generators. Wind power generation system. Power generation from biomass (gasification & Dendro thermal), Mini and micro small hydel plants.

UNIT-V

Fuel cells: Fuel cells and its associated parameters. Use of animal energy for different agricultural operation

References:

1. Garg H.P. 1990. Advances in Solar Energy Technology; D. Publishing Company, Tokyo.
2. Alan L: Farredbruch & R.H. Buse. 1983. Fundamentals of Solar Academic Press, London.
3. Bansal N.K., Kleemann M. & Meliss Michael. 1990. Renewable Energy Sources & Conversion Technology; Tata Mecgrow Publishing Company, New Delhi.
4. Rathore N. S., Kurchania A. K. & N.L. Panwar. 2007. Non Conventional Energy Sources, Himanshu Publications.
5. Mathur, A.N. & N.S. Rathore. 1992. Biogas Production Management & Utilization. Himanshu Publications, Udaipur.
6. Khandelwal, K.C. & S.S. Mahdi. 1990. Biogas Technology.
7. Rai, G.D. 2013. Non-Conventional Energy Sources, Khanna Publishers, Delhi.
8. Mathur A.N. & N.S. Rathore. Renewable Energy Sources Bohra Ganesh Publications, Udaipur.

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Semester – VI

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Computer Programming and Data Structures	BTA-5601	2-3 hr	3

Course Outcomes:

CO 1 Understand the high level language.

CO 2 Understand the algorithm of a problem and coding of that algorithm.

CO 3 Know all the details of data types, operators, library functions, decision making, branching, looping, arrays, user defined functions, recursion, string functions, structures and union, pointers, stacks, queues, linked lists.

Course Contents:

UNIT 1

Introduction to high level languages, Primary data types and user defined data types, Variables, typecasting, Operators, Building and evaluating expressions

UNIT 2

Managing input and output, Decision making, Branching, Looping, Arrays

UNIT 3

Standard library functions, User defined functions, passing arguments and returning values, recursion, scope and visibility of a variable, String functions

UNIT 4

Structures and union, Pointers

UNIT 5

Stacks, Push/Pop operations, Queues, Insertion and deletion operations, Linked lists.

References:

1. Rajaraman V. 1985. Computer Oriented Numerical Methods. Prentice Hall of India. Pvt. Ltd., New Delhi.
2. Balagurusamy E. 1990. Programming in ‘C’. Tata McGraw Hill Publishing Co. Ltd., 12/4 Asaf Ali Road, New Delhi.
3. Rajaraman V. 1995. Computer Programming in ‘C’. Prentice Hall of India Pvt.Ltd., New Delhi.

4.Bronson G and Menconi S. 1995. A First Book of ‘C’ Fundamentals of ‘C’ Programming. Jaico Publishing House, New Delhi
5.Sahni S.. Data Structures, Algorithms and Applications in C++. University press (India) Pvt Ltd / Orient Longman Pvt. Ltd.
6.Michael T. Goodrich, R. Tamassia and D Mount. Data structures and Algorithms in C++. Wiley Student Edition, John Wiley and Sons.
7.Mark Allen Weiss. Data Structures and Algorithm Analysis in C++. Pearson Education. Augenstein, Langsam and Tanenbaum. Data structures using C and C++. PHI/Pearson Education.
8.Drozdek Adam. Data Structures and Algorithms in C++. Vikas Publishing House / Thomson International Student Edition.

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Semester – VI

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Farm Machinery and Equipment-II	BTA-5602	2-3 hr	3

Course Outcomes:

CO 1 Know about the farm machineries used in agricultural production
CO 2 Know about construction, operation of different machines
CO 3 Know about the operating parameters and performance of the machines
CO 4 Know about the cost of operations and economics of the machines
CO 5 To solve numerical on different chapters

Course Contents:

UNIT-I
Plant protection equipment: sprayers and dusters. Classification of sprayers and sprays. Types of nozzles. Calculations for calibration of sprayers and chemical application rates.

UNIT-II
Intercultural equipment: Use of weeders – manual and powered. Study of functional requirements of weeders and main components. Familiarization of fertilizer application equipment. Study of harvesting operation – harvesting methods, harvesting terminology. Study of mowers – types, constructional details, working and adjustments. Study of shear type harvesting devices – cutter bar, inertial forces, counter balancing, terminology, cutting pattern.

UNIT-III
Reapers, binders and windrowers: Study of reapers, binders and windrowers – principle of operation and constructional details. Importance of hay conditioning, methods of hay conditioning, and calculation of moisture content of hay. Introduction to threshing systems – manual and mechanical systems. Types of threshing drums and their applications.

UNIT-IV
Thresher: Types of threshers- tangential and axial, their constructional details and cleaning systems. Study of factors affecting thresher performance. Study of grain combines, combine terminology, classification of grain combines, study of material flow in combines. Computation of combine losses, study of combine troubles and troubleshooting. Study of chaff cutters and capacity calculations. Study of straw combines – working principle and constructional details.

UNIT-V

Diggers and harvesting equipment: Study of root crop diggers – principle of operation, blade adjustment and approach angle, and calculation of material handled. Study of potato and groundnut diggers. Study of Cotton harvesting – Cotton harvesting mechanisms, study of cotton pickers and strippers, functional components. Study of maize harvesting combines. Introduction to vegetables and fruit harvesting equipment and tools.

References:

1. Kepner RA, Roy Barger & EL Barger. Principles of Farm Machinery.
2. Smith HP and LH Wilkey. Farm Machinery and Equipment.
3. Culpin Claude. Farm Machinery.
4. Srivastava AC. Elements of Farm Machinery.
5. Lal Radhey and AC Datta. Agricultural Engineering Principles of Farm Machinery.
6. Michael A M and Ojha, T.P. Principles of Agricultural Engineering. Jain Brothers, 873, East Park Road, Karol Bagh, New Delhi.
7. Jain S C, Philip Grace. Farm Machinery – An approach. Standard Publishers and Distributors, 1705-B, NaiSarak, Post Box No-1066, New Delhi-110006.

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Semester – VI

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Post-Harvest Engineering of Horticultural Crops	BTA-5603	2-3 hr	2

Course Outcomes:

CO 1 Know the different means of storage and value addition of fruits and vegetables along with the cold chain

CO 2 Know the different unit operations in processing of major horticultural crops of the country and state

CO 3 Understand the working principles of different machineries used for processing of fruits, vegetables and spices

CO 4 Understand the basics of selection of appropriate machines/ equipment for various applications of processing of horticultural crops

Course Contents:

UNIT-I

Introduction: Importance of processing of fruits and vegetables, spices, condiments and flowers. Characteristics and properties of horticultural crops important for processing, Peeling: Different peeling methods and devices (manual peeling, mechanical peeling, chemical peeling, and thermal peeling), Slicing of horticultural crops: equipment for slicing, shredding, crushing, chopping, juice extraction, etc., Blanching: Importance and objectives; blanching methods, effects on food (nutrition, colour, pigment, texture).

UNIT-II

Chilling and freezing: Application of refrigeration in different perishable food products, Thermophilic, mesophilic & Psychrophilic micro-organisms, Chilling requirements of different fruits and vegetables, Freezing of food, freezing time calculations, slow and fast freezing, Equipment for chilling and freezing (mechanical & cryogenic), Effect on food during chilling and freezing, Cold storage heat load calculations and cold storage design, refrigerated vehicle and cold chain system,

UNIT-III

Dryers, packaging and transportation: Dryers for fruits and vegetables, Osmo-dehydration, Packaging of horticultural commodities, Packaging requirements (in terms of light transmittance, heat, moisture and gas proof, microorganisms, mechanical strength), Different types of packaging materials commonly used for raw and processed fruits and vegetables products, bulk and retail packages and packaging machines, handling and transportation of fruits and vegetables, Pack house technology

UNIT-IV

Storage, preservation technology: Minimal processing, Common methods of storage, Low temperature storage, evaporative cooled storage, Controlled atmospheric storage, Modified atmospheric packaging, Preservation Technology, General methods of preservation of fruits and vegetables, Brief description and advantages and disadvantages of different physical/ chemical and other methods of preservation

UNIT-V

Post harvest management: Flowcharts for preparation of different finished products, Important parameters and equipment used for different unit operations, Post harvest management and equipment for spices and flowers, Quality control in Fruit and vegetable processing industry. Food supply chain.

References:

1. Arthey, D. and Ashurst, P. R. 1966. Fruit Processing. Chapman and Hall, New York.
2. Pantastico, E.C.B. 1975. Postharvest physiology, handling and utilization of tropical and subtropical fruits and vegetables AVI Pub. Co., New Delhi.
3. Pandey, R.H. 1997. Postharvest Technology of fruits and vegetables (Principles and practices). Saroj Prakashan, Allahabad.
4. Sudheer, K P. and Indira, V. 2007. Post Harvest Engineering of horticultural crops. New india Publishing House.
5. Lal Giridhari, Siddappa and Tondon. 2001. Preservation of fruits and vegetables. ICAR, New Delhi
6. Srivastava and Sanjeev Kumar. 2008. Fruit and vegetable preservation: principles and practices. Kalyani Publishers.

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Semester – VI

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Water Harvesting and Soil Conservation Structures	BTA-5604	2-3 hr	3

Course Outcomes:

CO 1 Know different soil erosion control structures and their requirements for better soil and moisture conservation in different situations.

CO 2 Understand the theory behind flow through soil conservation structures and use specific energy and momentum concepts to analyze flow problems.

CO 3 Prepare plan and design water harvesting and permanent soil and water conservation engineering structures with cost estimation.

CO 4 Design permanent soil and water conservation structures like drop spillway, chute spillway, drop inlet spillways, earthen embankments and water harvesting structures and estimate their cost.

CO 5 Recommend different structures for better soil and moisture conservation in problem areas.

Course Contents:

UNIT-I

Introduction: Water harvesting -principles, importance and issues. Water harvesting techniques - classification based on source, storage and use. Runoff harvesting – short-term and long-term techniques. Short-term harvesting techniques - terracing and bunding, rock and ground catchments.

UNIT-II

Farm ponds, construction and design criteria: Long-term harvesting techniques - purpose and design criteria. Structures - farm ponds - dug-out and embankment reservoir types, tanks and subsurface dykes. Farm pond - components, site selection, design criteria, capacity, embankment, mechanical and emergency spillways, cost estimation and construction.

UNIT-III

Structures for soil conservation and gully control: Percolation pond - site selection, design and construction details. Design considerations of *nala* bunds. Soil erosion control structures - introduction, classification and functional requirements. Permanent structures for soil conservation and gully control - check dams, drop, chute and drop inlet spillways - design requirements, planning for design, design procedures - hydrologic, hydraulic and structural design and stability analysis.

UNIT-IV

Hydraulic jump and drop spillway: Hydraulic jump and its application. Drop spillway - applicability, types - straight drop, box-type inlet spillways - description, functional use, advantages and disadvantages, straight apron and stilling basin outlet, structural components and functions. Loads on head wall, variables affecting equivalent fluid pressure, triangular load diagram for various flow conditions, creep line theory, uplift pressure estimation, safety against sliding, overturning, crushing and tension.

UNIT-V

Chute spillway, saf and drop inlet spillway: Chute spillway - description, components, energy dissipaters, design criteria of Saint Antony Falls (SAF) stilling basin and its limitations. Drop inlet spillway - description, functional use and design criteria.

References:

1. Singh Gurmel, C. Venkataraman, G. Sastry and B.P. Joshi. 1996. Manual of Soil and Water Conservation Practices. Oxford and IBH Publishing Co. Pvt. Ltd., New Delhi.
2. Michael, A.M. and T.P. Ojha. 2003. Principles of Agricultural Engineering. Volume II. 4th Edition, Jain Brothers, New Delhi.
3. Murthy, V.V.N. 2002. Land and Water Management Engineering. 4th Edition, Kalyani Publishers, New Delhi.
4. Schwab, G.O., D.D. Fangmeier, W.J. Elliot, R.K. Frevert. 1993. Soil and Water Conservation Engineering.4th Edition, John Wiley and Sons Inc. New York.
5. Suresh, R. 2014. Soil and Water Conservation Engineering. Standard Publisher Distributors, New Delhi.
6. Samra, J.S., V.N. Sharda and A.K. Sikka. 2002. Water Harvesting and Recycling: Indian Experiences. CSWCR&TI, Dehradun, Allied Printers, Dehradun.
7. Theib Y. Oweis, Dieter Prinz and Ahmed Y. Hachum. 2012. Rainwater Harvesting for Agriculture in the Dry Areas. CRC Press, Taylor and Francis Group, London.
8. Studer Rima Mekdaschi and Hanspeter Liniger. 2013. Water Harvesting - Guidelines to Good Practice. Centre for Development and Environment, University of Bern, Switzerland.
9. Chow, V.T.1985. Open-Channel Hydraulics. McGraw- Hill Book Company, Inc.
10. Sharda, V.N., Juyal, G.P., Prakash, C. and Joshi, B.P.2007. Training Manual: Soil Conservation & Watershed Management (Vol.-II) – CSWCRTI Publication, Dehradun.
11. USDA.1964. Engineering Hand Book on Drop Spillways (Section-11). USDA, Soil Conservation Service.

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Semester – VI

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Groundwater, Wells and Pumps	BTA-5605	2-3 hr	3

Course Outcomes:

CO 1 To understand the knowledge on occurrence, distribution and hydraulics of ground water in a region/area and characteristics/properties of the water bearing formations.
CO 2 Selection of points using different investigation methods for construction of wells.
CO 3 To acquaint with the equipments and different methods used for construction of wells suitable for different formations and the design of deep/ shallow wells.
CO 4 To have knowledge on different recharge methods for augmentation of ground water.
CO 5 To know the quality of ground water and its suitability for irrigation, drinking and industry purpose.
CO 6 To acquaint with different types of water lifting devices such as pumps, hydraulic ram, their adaptability and selection of required size.
CO 7 Selection of site for construction of wells.
CO 8 Efficient design of wells in the field condition.

Course Contents:

UNIT-I
Introduction: Occurrence and movement of ground water; aquifer and its types; classification of wells, fully penetrating tubewells and open wells, familiarization of various types of bore wells;

UNIT-II
Design of open wells and tubewell and drilling methods: Design of open wells; groundwater exploration techniques; methods of drilling of wells: percussion, rotary, reverse rotary; design of tubewell and gravel pack, installation of well screen, completion and development of well;

UNIT-III
Groundwater hydraulics: Determination of aquifer parameters by different method such as Theis, Jacob and Chow's, Theis recovery method; well interference, multiple well systems, estimation of ground water potential, quality of ground water; artificial groundwater recharge techniques;

UNIT-IV
Pumping systems: water lifting devices; different types of pumps, classification of pumps, component parts of centrifugal pumps, priming, pump selection, installation and trouble shooting,

UNIT-V

Characteristics of pumps: Effect of speed on capacity, head and power, effect of change of impeller dimensions on performance characteristics; hydraulic ram, propeller pumps, mixed flow pumps and their performance characteristics; deep well turbine pump and submersible pump.

References:

1. D.K. Todd, Ground water hydrology,. John Wiley & Sons, New York
2. S. P. Garg, Groundwater and tube wells,. Oxford & IBH Publishing Co.Ltd., New Delhi.
3. H.M.Raghunath, Ground water, new age publications, New Delhi
4. A. M. Michael, S. D. Khepar and S. K. Sondhi. Water well & Pump Engineering, Tata Mc-Graw Hill
5. R. Lal. Irrigation Hydraulics, Saroj Prakashan, Allahabad-02
6. H.S. Nagabhusan Ground water hydrology, CBS Publishers and Distributors, New Delhi.
7. J.C. Paul and B. Panigrahi. Practical Manual on Ground Water Hydrology, CAET, OUAT, Bhubaneswar.

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Tractor and Farm Machinery Operation and Maintenance	BTA-5606	2-3 hr	2

Course Outcomes:

CO 1 To enable the students to know the operation of tractor and farm machinery.

CO 2 To enable the students to understand the maintenance of tractor and farm machinery.

Course Contents:

Unit-I

Familiarization with different makes and models of agricultural tractors. Identification of functional systems including fuels system, cooling system, transmission system, steering and hydraulic systems. Study of maintenance points to be checked before starting a tractor.

Unit- II

Familiarization with controls on a tractor. Safety rules and precautions to be observed while driving a tractor. Driving practice of tractor. Practice of operating a tillage tool (mould-board plough/ disc plough) and their adjustment in the field. Study of field patterns while operating a tillage implement. Hitching & De-hitching of mounted and trail type implement to the tractor. Driving practice with a trail type trolley – forward and in reverse direction.

Unit-III

Introduction to tractor maintenance – precautionary and break-down maintenance. Tractor starting with low battery charge. Introduction to trouble shooting in tractors. Familiarization with tools for general and special maintenance.

Unit -IV

Introduction to scheduled maintenance after 10, 100, 300, 600, 900 and 1200 hours of operation. Safety hints. Top end overhauling. Fuel saving tips. Preparing the tractor for storage.

Unit -V

Care and maintenance procedure of agricultural machinery during operation and off-season. Repair and maintenance of implements – adjustment of functional parameters in tillage implements. Replacement of broken components in tillage implements. Replacement of furrow openers and change of blades of rotavators. Maintenance of cutter bar in a reaper. Adjustments in a thresher for different crops. Replacement of V-belts on implements. Setting of agricultural machinery workshop.

References:

1. Ghosh RK and S Swan. Practical Agricultural Engineering.
2. Black PO and WE Scahill. Diesel Engine Manual.
3. Southorn N. Tractor operation and maintenance.
4. Jain SC and CR Rai. Farm Tractor Maintenance and Repair.
5. Operators manuals of tractors.
6. Service manuals provided by manufacturers.

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Semester – VI

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Dairy and Food Engineering	BTA-5607	2-3 hr	3

Course Outcomes:

CO 1 Understand the different unit operations in processing and value addition of different dairy and food products.

CO 2 Understand the different types of equipment and their working principles used for processing and dairy and food products.

CO 3 Understand the basic principles of selection, operation and maintenance of different equipment for processing of milk and other food products for storage and value addition.

CO 4 Guide a prospective entrepreneur for basic layout and selection of equipment for a food processing plant.

Course Contents:

UNIT-I

Food preservation methods: Deterioration in food products and their controls, Physical, chemical and biological methods of food preservation. Nanotechnology: History, fundamental concepts, tools and techniques nanomaterials, applications in food packaging and products, implications, environmental impact of nanomaterials and their potential effects on global economics, regulation of nanotechnology.

UNIT-II

Milk properties: Dairy development in India, Engineering, thermal and chemical properties of milk and milk products, Process flow charts for product manufacture,

UNIT-III

Unit operation systems: Unit operation of various dairy and food processing systems. Principles and equipment related to receiving of milk, pasteurization, sterilization, homogenization, centrifugation and cream separation.

UNIT-IV

MILK PRODUCTS: Preparation methods and equipment for manufacture of cheese, paneer, butter and ice cream, Filling and packaging of milk and milk products; Dairy plant design and layout, Plant utilities; Principles of operation and equipment for thermal processing, Canning, Aseptic processing,

UNIT-V

Evaporation of food products: principle, types of evaporators, steam economy, multiple effect evaporation, vapour recompression, Drying of liquid and perishable foods: principles of drying, spray drying, drum drying, freeze drying, Filtration: principle, types of filters; Membrane separation, RO, Nano-filtration, Ultra filtration and Macro-filtration, equipment and applications, Non-thermal and other alternate thermal processing in Food processing.

References:

1. Ahmed, T. 1997. Dairy Plant Engineering and Management. 4th Ed. Kitab Mahal.
2. McCabe, W.L. and Smith, J. C. 1999. Unit Operations of Chemical Engineering. McGraw Hill.
3. Rao, D.G. Fundamentals of Food Engineering. PHI learning Pvt. Ltd. New Delhi.
4. Singh, R.P. & Heldman, D.R. 1993. Introduction to Food Engineering. Academic Press.
5. Toledo, R. T. 1997. Fundamentals of Food Process Engineering. CBS Publisher.
6. Dash, S K, Rayaguru K, Khan, M K. 2012. Concepts in Dairy and Food Engineering. OUAT, Bhubaneswar, 114 p.
7. Sahoo, N R, Dash S K, Pal, U S. 2012. Concepts in Food and Dairy Technology. OUAT, Bhubaneswar, 114 p.

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Semester – VI

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Bio-Energy Systems: Design and Applications	BTA-5608	2-3 hr	3

Course Outcomes:

CO 1 The different biomass sources, their properties and their applications for energy

CO 2 Production of biomass

CO 3 Thermo chemical and biochemical process of biomass for bioenergy and fuel production

Course Contents:

UNIT-I

Introduction: Fermentation processes and its general requirements, An overview of aerobic and anaerobic fermentation processes and their industrial application. Heat transfer processes in anaerobic digestion systems, land fill gas technology and potential.

UNIT-II

Biomass production: Wastelands, classification and their use through energy plantation, selection of species, methods of field preparation & transplanting. Harvesting of biomass & coppicing characteristics.

UNIT-III

Biomass preparation techniques: Biomass preparation techniques for harnessing (size reduction, densification and drying). Thermo-chemical degradation. History of small gas producer engine system. Chemistry of gasification.

UNIT-IV

Gas producer – type, operating principle. Gasifier fuels, properties, preparation, conditioning of producer gas, application, shaft power generation, thermal application and economics.

UNIT-V

Biodiesel production: Trans-esterification for biodiesel production. A range of bio-hydrogen production routes. Environmental aspect of bio-energy, assessment of greenhouse gas mitigation potential.

References:

1. British BioGen. 1997, Anaerobic digestion of farm and food processing practices- Good practice guidelines, London, available on www.britishbiogen.co.UK.
2. Butler, S. 2005. Renewable Energy Academy: Training wood energy professionals.
3. Centre for biomass energy. 1998. Straw for energy production; Technology- Environment-Ecology. Available: www.ens.dk.

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B. Tech. (AGRICULTURAL ENGINEERING)

New Scheme Based On ICAR Flexible Curriculum

Semester – VIII

Course Content

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
B. Tech (Agricultural Engineering)	Development of Processed Products	BTA-5801	2-3 hr	3

Course Outcomes:

CO 1 Unit operations and equipment used for different food processing operations.

CO 2 Processing technologies for value addition of cereals, pulses, oilseeds, vegetables, fruits, milk, fish, meat and poultry products.

CO 3 Know the operations and equipment used in food processing industries

CO 4 Have a basic understanding of process technologies for processing of cereals, pulses, oilseeds, vegetables, fruits, milk, fish, meat and poultry.

Course Contents:

UNIT-I

Introduction: Process design, Process flow chart with mass and energy balance, Unit operations and equipments for processing.

UNIT-II

Technology: New product development, Technology for value added products from cereal, pulses and oil seeds.

UNIT-III

Processing: Milling, puffing, flaking, Roasting, Bakery products, snack food. Extruded products, oil extraction and refining.

UNIT-IV

Technology for value added products: Technology for value added products from fruits, vegetables and spices, Canned foods, Frozen foods, dried and fried foods, Fruit juices, Sauce, Sugar based confection, Candy, Fermented food product, spice extracts.

UNIT-V

Technology for animal produce: Technology for animal produce processing , meat, poultry, fish, egg products, Health food, Nutra-ceuticals and functional food, Organic food.

References:

1. Geankoplis C. J. Transport processes and unit operations, Prentice-Hall.
2. Rao, D. G. Fundamentals of Food Engineering PHI Learning Pvt. Ltd, New Delhi.
3. Norman N. Potter and Joseph H. Hotchikss. Food Science. Chapman and Hall Pub.
4. Acharya, K T Everyday Indian Processed foods. National Book Trust.

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New Scheme Based On ICAR Flexible Curriculum

Semester – VIII

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Tractor Design and Testing	BTA-5802	2-3 hr	3

Course Outcomes:

CO 1 Procedure for design and development of agricultural tractor,

CO 2 Parameters for balanced design of tractor for stability & weight distribution,

CO 3 Traction theory, hydraulic lift and hitch system design.

CO 4 Design of mechanical power transmission in agricultural tractors: single disc, multi disc and cone clutches.

CO 5 Rolling friction and anti-friction bearings.

CO 6 Design of Ackerman Steering and tractor hydraulic steering.

CO 7 Study of special design features of tractor engines and their selection viz. cylinder, piston, piston pin, crankshaft, etc.

CO 8 Design of seat and controls of an agricultural tractor. Tractor Testing.

Course Contents:

UNIT-I

Introduction: Procedure for design and development of agricultural tractor, tractor testing.

UNIT-II

Parameters of tractor design: Study of parameters for balanced design of tractor for stability & weight distribution, traction theory, hydraulic lift and hitch system design.

UNIT-III

Mechanical power transmission: Design of mechanical power transmission in agricultural tractors: single disc, multi disc and cone clutches. Rolling friction and anti-friction bearings.

UNIT-IV

Features of tractor engines: Design of Ackerman steering & tractor hydraulic steering. Study of special design features of tractor engines & their selection: cylinder, piston, piston pin, crankshaft, etc.

UNIT-V

Design of seat and controls of an agricultural tractor.

References:

1. Liljedahl J B & Others. Tractors and Their Power Units.
2. Raymond N, EA Yong and S Nicolas. Vehicle Traction Mechanics.
3. Maleev VL. Internal Combustion Engines.

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New Scheme Based On ICAR Flexible Curriculum

Semester – VIII

Course Content

Branch	**Subject Title**	**Subject Code**	**Contact Hours per Week**	**Total Credits**
B. Tech (Agricultural Engineering)	Wasteland Development	BTA-5803	2-3 hr	3

Course Outcomes:

CO 1 To understand the concept of land degradation, classification and mapping of wastelands.

CO 2 To acquaint the students about planning of the wasteland development keeping in view agro climatic conditions, development options, contingency plans, conservation measures, water harvesting and recycling methods in consideration.

CO 3 To gain knowledge on different land reclamation and rehabilitation measures for wasteland development.

CO 4 To impart knowledge in the use of micro-irrigation for sustainable wasteland development against adverse situations like drought and water source situations.

Course Contents:

UNIT-I

Introduction: Land degradation – concept, classification - arid, semiarid, humid and sub-humid regions, denuded range land and marginal lands. Wastelands - factors causing, classification and mapping of wastelands, planning of wastelands development - constraints, agro-climatic conditions, development options, contingency plans.

UNIT-II

Conservation structures: gully stabilization, ravine rehabilitation, sand dune stabilization, water harvesting and recycling methods.

UNIT-III

Agricultural practices: Afforestation - agro-horti-forestry-silvipasture methods, forage and fuel crops - socioeconomic constraints. Shifting cultivation, optimal land use options.

UNIT-IV

Reclamation of wasteland: Wasteland development – hills, semi-arid, coastal areas, water scarce areas, reclamation of waterlogged and salt-affected lands. Mine spoils- impact, land degradation and reclamation and rehabilitation, slope stabilization and mine environment management.

UNIT-V

Benefit-cost analysis: Micro-irrigation in wastelands development. Sustainable wasteland development -drought situations, socio-economic perspectives. Government policies. Participatory approach. Preparation of proposal for wasteland development and benefit-cost analysis.

References:

1. Abrol I P. and V. V. Dhruvanarayana. 1998. Technologies for wasteland development. ICAR, New Delhi.
2. Ambast, S.K., S.K. Gupta and Gurcharan Singh (Eds.) 2007. Agricultural Land Drainage Reclamation of waterlogged saline lands. Central Soil Salinity Research Institute, Karnal, Haryana
3. Hridai Ram yadav. 2013. Management of Wastelands. Concept Publishing Company. New Delhi
4. Karthikeyan, C., K. Thangaraja, C. Cinthia Fernandez and K. Chandrakandon. 2009. Dry land Agriculture and Wasteland Management. Atlantic Publishers and distributors Pvt. Ltd. New delhi
5. Rattan Lal and B. A. Stewart(Ed). 2015. Soil Management of smallholder Agriculture, Volume21 of Advances in Soil Science. CRC Press, Taylor and Francis Group, Florida, USA.
6. Robert Malliva and Thomas Missimer. 2012. Arid Lands water evaluation and management. Springer Heidelberg New York.
7. Viranmani, S.M. (Ed.) 2010. Degraded and wastelands of India: Status and spatial Distribution. ICAR, New Delhi.
8. Swaminathan, M.S. 2010. Science and Integrated Rural Development. Concept Publishing Company (P) Ltd., Delhi.

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B. Tech. (AGRICULTURAL ENGINEERING)

Program Learning Outcomes: (PLOs)

1. Identification and sustainable design solutions to solve agricultural engineering problems applying the knowledge of basic and engineering sciences.
2. Ability to identify, formulate and solve agricultural engineering problems.
3. Pursue successful professional career in agro-industries, government organization, educational/ research/ extension institute adopting appropriate technology, resources and modelling.
4. Adapt in a world of evolving technology with professional ethics and take lead roles in development of agricultural engineering and allied enterprises for betterment of society.
5. knowledge of appropriate agricultural, and/or biological sciences, and/or natural resource topics
6. Competencies in relevant fields such as: farm machinery /farm mechanization systems, irrigation and drainage engineering systems, natural resource management systems, food processing systems, renewable energy systems and structural designs.
7. Understanding of professional and ethical responsibility.